

Power-Law Distributions in Empirical Data*

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Abstract. Power-law distributions occur in many situations of scientific interest and have significant consequences for our understanding of natural and man-made phenomena. Unfortunately, the detection and characterization of power laws is complicated by the large fluctuations that occur in the tail of the distribution—the part of the distribution representing large but rare events—and by the difficulty of identifying the range over which power-law behavior holds. Commonly used methods for analyzing power-law data, such as least-squares fitting, can produce substantially inaccurate estimates of parameters for power-law distributions, and even in cases where such methods return accurate answers they are still unsatisfactory because they give no indication of whether the data obey a power law at all. Here we present a principled statistical framework for discerning and quantifying power-law behavior in empirical data. Our approach combines maximum-likelihood fitting methods with goodness-of-fit tests based on the Kolmogorov–Smirnov (KS) statistic and likelihood ratios. We evaluate the effectiveness of the approach with tests on synthetic data and give critical comparisons to previous approaches. We also apply the proposed methods to twenty-four real-world data sets from a range of different disciplines, each of which has been conjectured to follow a power-law distribution. In some cases we find these conjectures to be consistent with the data, while in others the power law is ruled out.

Key words. power-law distributions, Pareto, Zipf, maximum likelihood, heavy-tailed distributions, likelihood ratio test, model selection

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1. Introduction. Many empirical quantities cluster around a typical value. The speeds of cars on a highway, the weights of apples in a store, air pressure, sea level, the temperature in New York at noon on a midsummer’s day: all of these things vary somewhat, but their distributions place a negligible amount of probability far from the typical value, making the typical value representative of most observations. For instance, it is a useful statement to say that an adult male American is about 180cm tall because no one deviates very far from this height. Even the largest deviations, which are exceptionally rare, are still only about a factor of two from the mean in

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either direction and hence the distribution can be well characterized by quoting just its mean and standard deviation.

Not all distributions fit this pattern, however, and while those that do not are often considered problematic or defective for just that reason, they are at the same time some of the most interesting of all scientific observations. The fact that they cannot be characterized as simply as other measurements is often a sign of complex underlying processes that merit further study.

Among such distributions, the *power law* has attracted particular attention over the years for its mathematical properties, which sometimes lead to surprising physical consequences, and for its appearance in a diverse range of natural and man-made phenomena. The populations of cities, the intensities of earthquakes, and the sizes of power outages, for example, are all thought to follow power-law distributions. Quantities such as these are not well characterized by their typical or average values. For instance, according to the 2000 U.S. Census, the average population of a city, town, or village in the United States is 8226. But this statement is not a useful one for most purposes because a significant fraction of the total population lives in cities (New York, Los Angeles, etc.) whose populations are larger by several orders of magnitude. Extensive discussions of this and other properties of power laws can be found in the reviews by Mitzenmacher [40], Newman [43], and Sornette [55], and references therein.

Mathematically, a quantity x obeys a power law if it is drawn from a probability distribution

$$(1.1) \quad p(x) \propto x^{-\alpha},$$

where α is a constant parameter of the distribution known as the *exponent* or *scaling parameter*. The scaling parameter typically lies in the range $2 < \alpha < 3$, although there are occasional exceptions.

In practice, few empirical phenomena obey power laws for all values of x . More often the power law applies only for values greater than some minimum $x_{\min}$. In such cases we say that the *tail* of the distribution follows a power law.

In this article we address a recurring issue in the scientific literature, the question of how to recognize a power law when we see one. In practice, we can rarely, if ever, be certain that an observed quantity is drawn from a power-law distribution. The most we can say is that our observations are consistent with the hypothesis that x is drawn from a distribution of the form of (1.1). In some cases we may also be able to rule out some other competing hypotheses. In this paper we describe in detail a set of statistical techniques that allow one to reach conclusions like these, as well as methods for calculating the parameters of power laws when we find them. Many of the methods we describe have been discussed previously; our goal here is to bring them together to create a complete procedure for the analysis of power-law data. A short description summarizing this procedure is given in Box 1. Software implementing it is also available online.¹

Practicing what we preach, we also apply our methods to a large number of data sets describing observations of real-world phenomena that have at one time or another been claimed to follow power laws. In the process, we demonstrate that several of them cannot reasonably be considered to follow power laws, while for others the power-law hypothesis appears to be a good one, or at least is not firmly ruled out.

¹See <http://www.santafe.edu/~aaronc/powerlaws/>.

Box 1: Recipe for analyzing power-law distributed data

This paper contains much technical detail. In broad outline, however, the recipe we propose for the analysis of power-law data is straightforward and goes as follows.

1. Estimate the parameters $x_{\min}$ and α of the power-law model using the methods described in section 3.
2. Calculate the goodness-of-fit between the data and the power law using the method described in section 4. If the resulting p -value is greater than 0.1, the power law is a plausible hypothesis for the data, otherwise it is rejected.
3. Compare the power law with alternative hypotheses via a likelihood ratio test, as described in section 5. For each alternative, if the calculated likelihood ratio is significantly different from zero, then its sign indicates whether or not the alternative is favored over the power-law model.

Step 3, the likelihood ratio test for alternative hypotheses, could in principle be replaced with any of several other established and statistically principled approaches for model comparison, such as a fully Bayesian approach [31], a cross-validation approach [58], or a minimum description length approach [20], although these methods are not described here.

2. Definitions. We begin our discussion of the analysis of power-law distributed data with some brief definitions of the basic quantities involved.

Power-law distributions come in two basic flavors: continuous distributions governing continuous real numbers and discrete distributions where the quantity of interest can take only a discrete set of values, typically positive integers.

Let x represent the quantity in whose distribution we are interested. A continuous power-law distribution is one described by a probability density $p(x)$ such that

$$(2.1) \quad p(x) dx = \Pr(x \leq X < x + dx) = Cx^{-\alpha} dx,$$

where X is the observed value and C is a normalization constant. Clearly this density diverges as $x \rightarrow 0$ so (2.1) cannot hold for all $x \geq 0$; there must be some lower bound to the power-law behavior. We will denote this bound by $x_{\min}$. Then, provided $\alpha > 1$, it is straightforward to calculate the normalizing constant and we find that

$$(2.2) \quad p(x) = \frac{\alpha - 1}{x_{\min}} \left(\frac{x}{x_{\min}} \right)^{-\alpha}.$$

In the discrete case, x can take only a discrete set of values. In this paper we consider only the case of integer values with a probability distribution of the form

$$(2.3) \quad p(x) = \Pr(X = x) = Cx^{-\alpha}.$$

Again this distribution diverges at zero, so there must be a lower bound $x_{\min} > 0$ on the power-law behavior. Calculating the normalizing constant, we then find that

$$(2.4) \quad p(x) = \frac{x^{-\alpha}}{\zeta(\alpha, x_{\min})},$$

Table 1 Definition of the power-law distribution and several other common statistical distributions. For each distribution we give the basic functional form $f(x)$ and the appropriate normalization constant C such that $\int_{x_{\min}}^{\infty} C f(x) dx = 1$ for the continuous case or $\sum_{x=x_{\min}}^{\infty} C f(x) = 1$ for the discrete case.

	Name	Distribution $p(x) = C f(x)$	
		$f(x)$	C
Continuous	Power law	$x^{-\alpha}$	$(\alpha - 1)x_{\min}^{\alpha-1}$
	Power law with cutoff	$x^{-\alpha}e^{-\lambda x}$	$\frac{\lambda^{1-\alpha}}{\Gamma(1-\alpha, \lambda x_{\min})}$
	Exponential	$e^{-\lambda x}$	$\lambda e^{\lambda x_{\min}}$
	Stretched exponential	$x^{\beta-1}e^{-\lambda x^{\beta}}$	$\beta \lambda e^{\lambda x_{\min}^{\beta}}$
	Log-normal	$\frac{1}{x} \exp\left[-\frac{(\ln x - \mu)^2}{2\sigma^2}\right]$	$\sqrt{\frac{2}{\pi\sigma^2}} \left[\operatorname{erfc}\left(\frac{\ln x_{\min} - \mu}{\sqrt{2}\sigma}\right) \right]^{-1}$
Discrete	Power law	$x^{-\alpha}$	$1/\zeta(\alpha, x_{\min})$
	Yule distribution	$\frac{\Gamma(x)}{\Gamma(x+\alpha)}$	$(\alpha - 1) \frac{\Gamma(x_{\min} + \alpha - 1)}{\Gamma(x_{\min})}$
	Exponential	$e^{-\lambda x}$	$(1 - e^{-\lambda}) e^{\lambda x_{\min}}$
	Poisson	$\mu^x / x!$	$\left[e^{\mu} - \sum_{k=0}^{x_{\min}-1} \frac{\mu^k}{k!} \right]^{-1}$

where

$$(2.5) \quad \zeta(\alpha, x_{\min}) = \sum_{n=0}^{\infty} (n + x_{\min})^{-\alpha}$$

is the generalized or Hurwitz zeta function. Table 1 summarizes the basic functional forms and normalization constants for these and several other distributions that will be useful.

In many cases it is useful to consider also the complementary cumulative distribution function or CDF of a power-law distributed variable, which we denote $P(x)$ and which for both continuous and discrete cases is defined to be $P(x) = \Pr(X \geq x)$. For instance, in the continuous case,

$$(2.6) \quad P(x) = \int_x^{\infty} p(x') dx' = \left(\frac{x}{x_{\min}} \right)^{-\alpha+1}.$$

In the discrete case,

$$(2.7) \quad P(x) = \frac{\zeta(\alpha, x)}{\zeta(\alpha, x_{\min})}.$$

Because formulas for continuous distributions, such as (2.2), tend to be simpler than those for discrete distributions, it is common to approximate discrete power-law behavior with its continuous counterpart for the sake of mathematical convenience. But a word of caution is in order: there are several different ways to approximate a discrete power law by a continuous one and though some of them give reasonable results, others do not. One relatively reliable method is to treat an integer power law as if the values of x were generated from a continuous power law then rounded to the nearest integer. This approach gives quite accurate results in many applications. Other approximations, however, such as truncating (rounding down) or simply

assuming that the probabilities of generation of integer values in the discrete and continuous cases are proportional, give poor results and should be avoided.

Where appropriate we will discuss the use of continuous approximations for the discrete power law in the sections that follow, particularly in section 3 on the estimation of best-fit values for the scaling parameter from observational data and in Appendix D on the generation of power-law distributed random numbers.

3. Fitting Power Laws to Empirical Data. We turn now to the first of the main goals of this paper, the correct fitting of power-law forms to empirical distributions. Studies of empirical distributions that follow power laws usually give some estimate of the scaling parameter α and occasionally also of the lower bound on the scaling region $x_{\min}$. The tool most often used for this task is the simple histogram. Taking the logarithm of both sides of (1.1), we see that the power-law distribution obeys $\ln p(x) = \alpha \ln x + \text{constant}$, implying that it follows a straight line on a doubly logarithmic plot. A common way to probe for power-law behavior, therefore, is to measure the quantity of interest x , construct a histogram representing its frequency distribution, and plot that histogram on doubly logarithmic axes. If in so doing one discovers a distribution that falls approximately on a straight line, then one can, if feeling particularly bold, assert that the distribution follows a power law, with a scaling parameter α given by the absolute slope of the straight line. Typically this slope is extracted by performing a least-squares linear regression on the logarithm of the histogram. This procedure dates back to Pareto's work on the distribution of wealth at the close of the 19th century [6].

Unfortunately, this method and other variations on the same theme generate significant systematic errors under relatively common conditions, as discussed in Appendix A, and as a consequence the results they give cannot be trusted. In this section we describe a generally accurate method for estimating the parameters of a power-law distribution. In section 4 we study the equally important question of how to determine whether a given data set really does follow a power law at all.

3.1. Estimating the Scaling Parameter. First, let us consider the estimation of the scaling parameter α . Estimating α correctly requires, as we will see, a value for the lower bound $x_{\min}$ of power-law behavior in the data. For the moment, let us assume that this value is known. In cases where it is unknown, we can estimate it from the data as well, and we will consider methods for doing this in section 3.3.

The method of choice for fitting parametrized models such as power-law distributions to observed data is the *method of maximum likelihood*, which provably gives accurate parameter estimates in the limit of large sample size [63, 7]. Assuming that our data are drawn from a distribution that follows a power law exactly for $x \geq x_{\min}$, we can derive maximum likelihood estimators (MLEs) of the scaling parameter for both the discrete and continuous cases. Details of the derivations are given in Appendix B; here our focus is on their use.

The MLE for the continuous case is [42]

$$(3.1) \quad \hat{\alpha} = 1 + n \left[\sum_{i=1}^n \ln \frac{x_i}{x_{\min}} \right]^{-1},$$

where x_i , $i = 1, \dots, n$, are the observed values of x such that $x_i \geq x_{\min}$. Here and elsewhere we use “hatted” symbols such as $\hat{\alpha}$ to denote estimates derived from data; hatless symbols denote the true values, which are often unknown in practice.

Equation (3.1) is equivalent to the well-known Hill estimator [24], which is known to be asymptotically normal [22] and consistent [37] (i.e., $\hat{\alpha} \rightarrow \alpha$ in the limit of large n). The standard error on $\hat{\alpha}$, which is derived from the width of the likelihood maximum, is

$$(3.2) \quad \sigma = \frac{\hat{\alpha} - 1}{\sqrt{n}} + O(1/n),$$

where the higher-order correction is positive; see Appendix B of this paper or any of the references [42], [43], or [66].

(We assume in these calculations that $\alpha > 1$, since distributions with $\alpha \leq 1$ are not normalizable and hence cannot occur in nature. It is possible for a probability distribution to go as $x^{-\alpha}$ with $\alpha \leq 1$ if the range of x is bounded above by some cutoff, but different MLEs are needed to fit such a distribution.)

The MLE for the case where x is a discrete integer variable is less straightforward. Reference [51] and more recently [19] treated the special case $x_{\min} = 1$, showing that the appropriate estimator for α is given by the solution to the transcendental equation

$$(3.3) \quad \frac{\zeta'(\hat{\alpha})}{\zeta(\hat{\alpha})} = -\frac{1}{n} \sum_{i=1}^n \ln x_i.$$

When $x_{\min} > 1$, a similar equation holds, but with the zeta functions replaced by generalized zetas [6, 8, 11],

$$(3.4) \quad \frac{\zeta'(\hat{\alpha}, x_{\min})}{\zeta(\hat{\alpha}, x_{\min})} = -\frac{1}{n} \sum_{i=1}^n \ln x_i,$$

where the prime denotes differentiation with respect to the first argument. In practice, evaluation of $\hat{\alpha}$ requires us to solve this equation numerically. Alternatively, one can estimate α by direct numerical maximization of the likelihood function itself, or equivalently of its logarithm (which is usually simpler):

$$(3.5) \quad \mathcal{L}(\alpha) = -n \ln \zeta(\alpha, x_{\min}) - \alpha \sum_{i=1}^n \ln x_i.$$

To find an estimate for the standard error on $\hat{\alpha}$ in the discrete case, we make a quadratic approximation to the log-likelihood at its maximum and take the standard deviation of the resulting Gaussian form for the likelihood as our error estimate (an approach justified by general theorems on the large-sample-size behavior of maximum likelihood estimates—see, for example, Theorem B.3 of Appendix B). The result is

$$(3.6) \quad \sigma = \frac{1}{\sqrt{n \left[\frac{\zeta''(\hat{\alpha}, x_{\min})}{\zeta(\hat{\alpha}, x_{\min})} - \left(\frac{\zeta'(\hat{\alpha}, x_{\min})}{\zeta(\hat{\alpha}, x_{\min})} \right)^2 \right]}},$$

which is straightforward to evaluate once we have $\hat{\alpha}$. Alternatively, (3.2) yields roughly similar results for reasonably large n and $x_{\min}$.

Although there is no exact closed-form expression for $\hat{\alpha}$ in the discrete case, an approximate expression can be derived using the approach mentioned in section 2 in which true power-law distributed integers are approximated as continuous reals rounded to the nearest integer. The details of the derivation are given in Appendix B.

The result is

$$(3.7) \quad \hat{\alpha} \simeq 1 + n \left[\sum_{i=1}^n \ln \frac{x_i}{x_{\min} - \frac{1}{2}} \right]^{-1}.$$

This expression is considerably easier to evaluate than the exact discrete MLE and can be useful in cases where high accuracy is not needed. The size of the bias introduced by the approximation is discussed in Appendix B. In practice, this estimator gives quite good results; in our own experiments we have found it to give results accurate to about 1% or better provided $x_{\min} \gtrsim 6$. An estimate of the statistical error on $\hat{\alpha}$ (which is quite separate from the systematic error introduced by the approximation) can be calculated by employing (3.2) again.

Another approach taken by some authors is simply to pretend that discrete data are in fact continuous and then use the MLE for continuous data, (3.1), to calculate $\hat{\alpha}$. This approach, however, gives significantly less accurate values of $\hat{\alpha}$ than (3.7) and, given that it is no easier to implement, we see no reason to use it in any circumstances.²

3.2. Performance of Scaling Parameter Estimators. To demonstrate the working of the estimators described above, we now test their ability to extract the known scaling parameters of synthetic power-law data. Note that in practical situations we usually do not know a priori, as we do in the calculations of this section, that our data are power-law distributed. In that case, our MLEs will give us no warning that our fits are wrong: they tell us only the best fit to the power-law form, not whether the power law is in fact a good model for the data. Other methods are needed to address the latter question, and are discussed in sections 4 and 5.

Using methods described in Appendix D, we have generated two sets of power-law distributed data, one continuous and one discrete, with $\alpha = 2.5$, $x_{\min} = 1$, and $n = 10\,000$ in each case. Applying our MLEs to these data we calculate that $\hat{\alpha} = 2.50(2)$ for the continuous case and $\hat{\alpha} = 2.49(2)$ for the discrete case. (Values in parentheses indicate the uncertainty in the final digit, calculated from (3.2) and (3.6).) These estimates agree well with the known true scaling parameter from which the data were generated. Figure 1 shows the distributions of the two data sets along with fits using the estimated parameters. (In this and all subsequent such plots, we show not the probability density function (PDF), but the complementary CDF $P(x)$. Generally, the visual form of the CDF is more robust than that of the PDF against fluctuations due to finite sample sizes, particularly in the tail of the distribution.)

In Table 2 we compare the results given by the MLEs to estimates of the scaling parameter made using several alternative methods based on linear regression: a straight-line fit to the slope of a log-transformed histogram, a fit to the slope of a histogram with “logarithmic bins” (bins whose width increases in proportion to x , thereby reducing fluctuations in the tail of the histogram), a fit to the slope of the CDF calculated with constant width bins, and a fit to the slope of the CDF calculated without any bins (also called a “rank-frequency plot”—see [43]). As the table shows, the MLEs give the best results, while the regression methods all give significantly biased values, except perhaps for the fits to the CDF, which produce biased estimates in the discrete case but do reasonably well in the continuous case. Moreover, in each

²The error involved can be shown to decay as $O(x_{\min}^{-1})$, while the error on (3.7) decays much faster, as $O(x_{\min}^{-2})$. In our own experiments we have found that for typical values of α we need $x_{\min} \gtrsim 100$ before (3.1) becomes accurate to about 1%, as compared to $x_{\min} \gtrsim 6$ for (3.7).

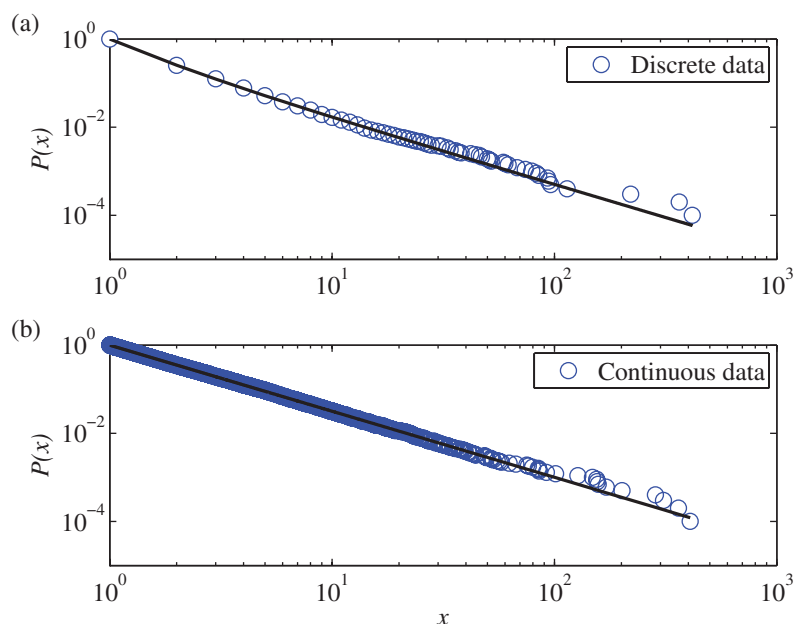


Fig. 1 Points represent the CDFs $P(x)$ for synthetic data sets distributed according to (a) a discrete power law and (b) a continuous power law, both with $\alpha = 2.5$ and $x_{\min} = 1$. Solid lines represent best fits to the data using the methods described in the text.

Table 2 Estimates of the scaling parameter α using various estimators for discrete and continuous synthetic data with $\alpha = 2.5$, $x_{\min} = 1$, and $n = 10\,000$ data points. LS denotes a least-squares fit to the logarithm of the probability. For the continuous data, the PDF was computed in two different ways, using bins of constant width 0.1 and using up to 500 bins of exponentially increasing width (so-called “logarithmic binning”). The CDF was also calculated in two ways, as the cumulation of the fixed-width histogram and as a standard rank-frequency function. In applying the discrete MLE to the continuous data, the noninteger part of each measurement was discarded. Accurate estimates are shown in **bold**.

Method	Notes	est. α (Discrete)	est. α (Continuous)
LS + PDF	const. width	1.5(1)	1.39(5)
LS + CDF	const. width	2.37(2)	2.480(4)
LS + PDF	log. width	1.5(1)	1.19(2)
LS + CDF	rank-freq.	2.570(6)	2.4869(3)
cont. MLE	–	4.46(3)	2.50(2)
disc. MLE	–	2.49(2)	2.19(1)

case where the estimate is biased, the corresponding error estimate gives no warning of the bias: there is nothing to alert unwary experimenters to the fact that their results are substantially incorrect. Figure 2 extends these results graphically by showing how the estimators fare as a function of the true α for a large selection of synthetic data sets with $n = 10\,000$ observations each.

Finally, we note that the MLEs are only guaranteed to be unbiased in the asymptotic limit of large sample size, $n \rightarrow \infty$. For finite data sets, biases are present but decay as $O(n^{-1})$ for any choice of $x_{\min}$ (see Appendix B and Figure 10). For very small data sets, such biases can be significant but in most practical situations

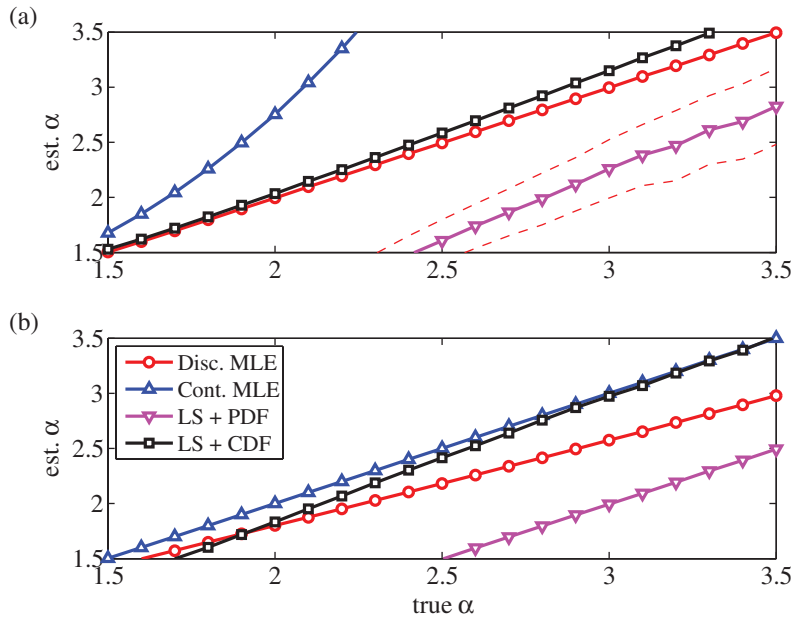


Fig. 2 Values of the scaling parameter estimated using four of the methods of Table 2 (we omit the methods based on logarithmic bins for the PDF and constant width bins for the CDF) for $n = 10\,000$ observations drawn from (a) discrete and (b) continuous power-law distributions with $x_{\min} = 1$. We omit error bars where they are smaller than the symbol size. Clearly, only the discrete MLE is accurate for discrete data, and the continuous MLE for continuous data.

they can be ignored because they are much smaller than the statistical error of the estimator, which decays as $O(n^{-1/2})$. Our experience suggests that $n \gtrsim 50$ is a reasonable rule of thumb for extracting reliable parameter estimates. For the examples shown in Figure 10 this gives estimates of α accurate to about 1%. Data sets smaller than this should be treated with caution. Note, however, that there are more important reasons to treat small data sets with caution. Namely, it is difficult to rule out alternative fits to such data, even when they are truly power-law distributed, and conversely the power-law form may appear to be a good fit even when the data are drawn from a non-power-law distribution. We address these issues in sections 4 and 5.

3.3. Estimating the Lower Bound on Power-Law Behavior. As we have said above it is normally the case that empirical data, if they follow a power-law distribution at all, do so only for values of x above some lower bound $x_{\min}$. Before calculating our estimate of the scaling parameter α , therefore, we need to first discard all samples below this point so that we are left with only those for which the power-law model is valid. Thus, if we wish our estimate of α to be accurate, we will also need an accurate method for estimating $x_{\min}$. If we choose too low a value for $x_{\min}$, we will get a biased estimate of the scaling parameter since we will be attempting to fit a power-law model to non-power-law data. On the other hand, if we choose too high a value for $x_{\min}$, we are effectively throwing away legitimate data points $x_i < \hat{x}_{\min}$, which increases both the statistical error on the scaling parameter and the bias from finite size effects.

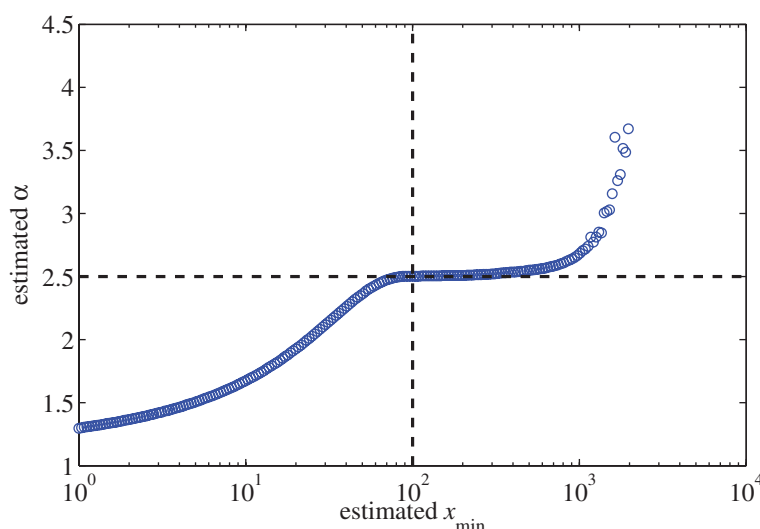


Fig. 3 Mean of the MLE for the scaling parameter for 5000 samples drawn from the test distribution, (3.10), with $\alpha = 2.5$, $x_{\min} = 100$, and $n = 2500$, plotted as a function of the value assumed for $x_{\min}$. Statistical errors are smaller than the data points in all cases.

The importance of using the correct value for $x_{\min}$ is demonstrated in Figure 3, which shows the maximum likelihood value $\hat{\alpha}$ of the scaling parameter averaged over 5000 data sets of $n = 2500$ samples, each drawn from the continuous form of (3.10) with $\alpha = 2.5$, as a function of the assumed value of $x_{\min}$, where the true value is 100. As the figure shows, the MLE gives accurate answers when $x_{\min}$ is chosen exactly equal to the true value, but deviates rapidly below this point (because the distribution deviates from power law) and more slowly above (because of dwindling sample size). It would probably be acceptable in this case for $x_{\min}$ to err a little on the high side (though not too much), but estimates that are too low could have severe consequences.

The most common ways of choosing $\hat{x}_{\min}$ are either to estimate visually the point beyond which the PDF or CDF of the distribution becomes roughly straight on a log-log plot, or to plot $\hat{\alpha}$ (or a related quantity) as a function of $\hat{x}_{\min}$ and identify a point beyond which the value appears relatively stable. But these approaches are clearly subjective and can be sensitive to noise or fluctuations in the tail of the distribution—see [57] and references therein. A more objective and principled approach is desirable. Here we review two such methods, one that is specific to discrete data and is based on a so-called marginal likelihood, and one that works for either discrete or continuous data and is based on minimizing the “distance” between the power-law model and the empirical data.

The first approach, put forward by Handcock and Jones [23], uses a generalized model to represent all of the observed data, both above and below $\hat{x}_{\min}$. Above $\hat{x}_{\min}$ the data are modeled by the standard discrete power-law distribution of (2.4); below $\hat{x}_{\min}$ each of the $\hat{x}_{\min} - 1$ discrete values of x are modeled by a separate probability $p_k = \Pr(X = k)$ for $1 \leq k < \hat{x}_{\min}$ (or whatever range is appropriate for the problem at hand). The MLE for p_k is simply the fraction of observations with value k . The task then is to find the value for $\hat{x}_{\min}$ such that this model best fits the observed data. One

cannot, however, fit such a model to the data directly within the maximum likelihood framework because the number of model parameters is not fixed: it is equal to $x_{\min}$.³ In this kind of situation, one can always achieve a higher likelihood by increasing the number of parameters, thus making the model more flexible, so the maximum likelihood is always achieved for $x_{\min} \rightarrow \infty$. A standard (Bayesian) approach in such cases is instead to maximize the *marginal likelihood* (also called the *evidence*) [29, 34], i.e., the likelihood of the data given the number of model parameters, integrated over the parameters' possible values. Unfortunately, the integral cannot usually be performed analytically, but one can employ a Laplace or steepest-descent approximation in which the log-likelihood is expanded to leading (i.e., quadratic) order about its maximum and the resulting Gaussian integral is carried out to yield an expression in terms of the value at the maximum and the determinant of the appropriate Hessian matrix [60]. Schwarz [50] showed that the terms involving the Hessian can be simplified for large n yielding an approximation to the log marginal likelihood of the form

$$(3.8) \quad \ln \Pr(x|x_{\min}) \simeq \mathcal{L} - \frac{1}{2}x_{\min} \ln n,$$

where $\mathcal{L}$ is the value of the conventional log-likelihood at its maximum. This type of approximation is known as a *Bayesian information criterion* or BIC. The maximum of the BIC with respect to $x_{\min}$ then gives the estimated value $\hat{x}_{\min}$.⁴

This method works well under some circumstances, but can also present difficulties. In particular, the assumption that $x_{\min} - 1$ parameters are needed to model the data below $x_{\min}$ may be excessive: in many cases the distribution below $x_{\min}$, while not following a power law, can nonetheless be represented well by a model with a much smaller number of parameters. In this case, the BIC tends to underestimate the value of $x_{\min}$ and this could result in biases on the subsequently calculated value of the scaling parameter. More importantly, it is also unclear how the BIC (and similar methods) can be generalized to the case of continuous data, for which there is no obvious choice for how many parameters are needed to represent the empirical distribution below $x_{\min}$.

Our second approach for estimating $x_{\min}$, proposed by Clauset, Young, and Gleditsch [11], can be applied to both discrete and continuous data. The fundamental idea behind this method is simple: we choose the value of $\hat{x}_{\min}$ that makes the probability distributions of the measured data and the best-fit power-law model as similar as possible above $\hat{x}_{\min}$. In general, if we choose $\hat{x}_{\min}$ higher than the true value $x_{\min}$, then we are effectively reducing the size of our data set, which will make the probability distributions a poorer match because of statistical fluctuation. Conversely, if we choose $\hat{x}_{\min}$ smaller than the true $x_{\min}$, the distributions will differ because of the fundamental difference between the data and model by which we are describing it. In between lies our best estimate.

There are a variety of measures for quantifying the distance between two probability distributions, but for nonnormal data the commonest is the Kolmogorov–Smirnov or KS statistic [46], which is simply the maximum distance between the CDFs of the

³There is one parameter for each of the p_k plus the scaling parameter of the power law. The normalization constant does not count as a parameter because it is fixed once the values of the other parameters are chosen, and $x_{\min}$ does not count as a parameter because we know its value automatically once we are given a list of the other parameters—it is just the length of that list.

⁴The same procedure of reducing the likelihood by $\frac{1}{2} \ln n$ times the number of model parameters to avoid overfitting can also be justified on non-Bayesian grounds for many model selection problems.

data and the fitted model:

$$(3.9) \quad D = \max_{x \geq x_{\min}} |S(x) - P(x)|.$$

Here $S(x)$ is the CDF of the data for the observations with value at least $x_{\min}$, and $P(x)$ is the CDF for the power-law model that best fits the data in the region $x \geq x_{\min}$. Our estimate $\hat{x}_{\min}$ is then the value of $x_{\min}$ that minimizes D .⁵

There is good reason to expect this method to produce reasonable results. Note in particular that for right-skewed data of the kind we consider here the method is especially sensitive to slight deviations of the data from the power-law model around $x_{\min}$ because most of the data, and hence most of the dynamic range of the CDF, lie in this region. In practice, as we show in the following section, the method appears to give excellent results and generally performs better than the BIC approach.

3.4. Tests of Estimates for the Lower Bound. As with our MLEs for the scaling parameter, we test our two methods for estimating $x_{\min}$ by generating synthetic data and examining the methods' ability to recover the known value of $x_{\min}$. For the tests presented here we use synthetic data drawn from a distribution with the form

$$(3.10) \quad p(x) = \begin{cases} C(x/x_{\min})^{-\alpha} & \text{for } x \geq x_{\min}, \\ Ce^{-\alpha(x/x_{\min}-1)} & \text{for } x < x_{\min}, \end{cases}$$

with $\alpha = 2.5$. This distribution follows a power law at $x_{\min}$ and above but an exponential below. Furthermore, it has a continuous slope at $x_{\min}$ and thus deviates only gently from the power law as we pass below this point, making for a challenging test. Figure 4a shows a family of curves from this distribution for different values of $x_{\min}$.

In Figure 4b we show the results of the application of both the BIC and KS methods for estimating $x_{\min}$ to a large collection of data sets drawn from (3.10). The plot shows the average estimated value $\hat{x}_{\min}$ as a function of the true $x_{\min}$ for the discrete case. The KS method appears to give good estimates of $x_{\min}$ in this case and performance is similar for continuous data also (not shown), although the results tend to be slightly more conservative (i.e., to yield slightly larger estimates $\hat{x}_{\min}$). The BIC method also performs reasonably, but, as the figure shows, the method displays a tendency to underestimate $x_{\min}$, as we might expect given the arguments of the previous section. Based on these observations, we recommend the KS method for estimating $x_{\min}$ for general applications.

These tests used synthetic data sets of $n = 50\,000$ observations, but good estimates of $x_{\min}$ can be extracted from significantly smaller data sets using the KS method; results are sensitive principally to the number n_{tail} of observations in the power-law part of the distribution. For both the continuous and discrete cases we find that good results can be achieved provided we have about 1000 or more observations in this part of the distribution. This figure does depend on the particular form of the non-power-law part of the distribution. In the present test, the distribution was designed specifically to make the determination of $x_{\min}$ challenging. Had we chosen a form that makes a more pronounced departure from the power law below

⁵We note in passing that this approach can easily be generalized to the problem of estimating a lower cut-off for data following other (non-power-law) types of distributions.

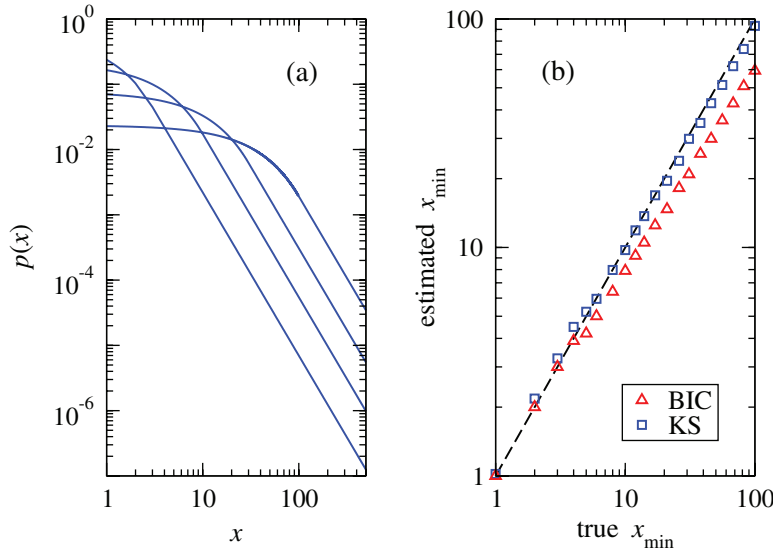


Fig. 4 (a) Examples of the test distribution, (3.10), used in the calculations described in the text, with power-law behavior for x above $x_{\min}$ but non-power-law behavior below. (b) The value of $x_{\min}$ estimated using the BIC and KS approaches as described in the text, plotted as a function of the true value for discrete data with $n = 50\,000$. Results are similar for continuous data.

$x_{\min}$, then the task of estimating $\hat{x}_{\min}$, would have been easier and presumably fewer observations would have been needed to achieve results of similar quality.

For some possible distributions there is, in a sense, no true value of $x_{\min}$. The distribution $p(x) = C(x + k)^{-\alpha}$ follows a power law in the limit of large x , but there is no value of $x_{\min}$ above which it follows a power law exactly. Nonetheless, in cases such as this, we would like our method to return an $\hat{x}_{\min}$ such that when we subsequently calculate a best-fit value for α we get an accurate estimate of the true scaling parameter. In tests with such distributions we find that the KS method yields estimates of α that appear to be asymptotically consistent, meaning that $\hat{\alpha} \rightarrow \alpha$ as $n \rightarrow \infty$. Thus again the method appears to work well, although it remains an open question whether one can derive rigorous performance guarantees.

Variations on the KS method are possible that use some other goodness-of-fit measure that may perform better than the KS statistic under certain circumstances. The KS statistic is, for instance, known to be relatively insensitive to differences between distributions at the extreme limits of the range of x because in these limits the CDFs necessarily tend to zero and one. It can be reweighted to avoid this problem and be uniformly sensitive across the range [46]; the appropriate reweighting is

$$(3.11) \quad D^* = \max_{x \geq \hat{x}_{\min}} \frac{|S(x) - P(x)|}{\sqrt{P(x)(1 - P(x))}}.$$

In addition, a number of other goodness-of-fit statistics have been proposed and are in common use, such as the Kuiper and Anderson–Darling statistics [13]. We have performed tests with each of these alternative statistics and find that results for the reweighted KS and Kuiper statistics are very similar to those for the standard KS statistic. The Anderson–Darling statistic, on the other hand, we find to be highly

conservative in this application, giving estimates $\hat{x}_{\min}$ that are too large by an order of magnitude or more. When there are many samples in the tail of the distribution, this degree of conservatism may be acceptable, but in most cases the reduction in the number of tail observations greatly increases the statistical error on our MLE for the scaling parameter and also reduces our ability to validate the power-law model.

Finally, as with our estimate of the scaling parameter, we would like to quantify the uncertainty in our estimate for $x_{\min}$. One way to do this is to make use of a nonparametric “bootstrap” method [16]. Given our n measurements, we generate a synthetic data set with a similar distribution to the original by drawing a new sequence of points x_i , $i = 1, \dots, n$, uniformly at random from the original data (with replacement). Using either method described above, we then estimate $x_{\min}$ and α for this surrogate data set. By taking the standard deviation of these estimates over a large number of repetitions of this process (say, 1000), we can derive principled estimates of our uncertainty in the original estimated parameters.

3.5. Other Techniques. We would be remiss should we fail to mention some of the other techniques in use for the analysis of power-law distributions, particularly those developed within the statistics and finance communities, where the study of these distributions has, perhaps, the longest history. We give only a brief summary of this material here; readers interested in pursuing the topic further are encouraged to consult the books by Adler, Feldman, and Taqqu [4] and Resnick [48] for a more thorough explanation.⁶

In the statistical literature, researchers often consider a family of distributions of the form

$$(3.12) \quad p(x) \propto L(x) x^{-\alpha},$$

where $L(x)$ is some slowly varying function, so that, in the limit of large x , $L(cx)/L(x) \rightarrow 1$ for any $c > 0$. An important issue in this case—as in the calculations presented in this paper—is finding the point $x_{\min}$ at which the $x^{-\alpha}$ can be considered to dominate over the nonasymptotic behavior of the function $L(x)$, a task that can be tricky if the data span only a limited dynamic range or if the non-power-law behavior $|L(x) - L(\infty)|$ decays only a little faster than $x^{-\alpha}$. In such cases, a visual approach—plotting an estimate $\hat{\alpha}$ of the scaling parameter as a function of $x_{\min}$ (called a Hill plot) and choosing for $\hat{x}_{\min}$ the value beyond which $\hat{\alpha}$ appears stable—is a common technique. Plotting other statistics, however, can often yield better results—see, for example, [33] and [57]. An alternative approach, quite common in the quantitative finance literature, is simply to limit the analysis to the largest observed samples only, such as the largest $\sqrt{n}$ or $\frac{1}{10}n$ observations [17].

The methods described in section 3.3, however, offer several advantages over these techniques. In particular, the KS method of section 3.3 gives estimates of $x_{\min}$ as least as good while being simple to implement and having low enough computational costs that it can be effectively used as a foundation for further analyses such as the calculation of p -values in section 4. And, perhaps more importantly, because the KS method removes the non-power-law portion of the data entirely from the estimation

⁶Another related area of study is “extreme value theory,” which concerns itself with the distribution of the largest or smallest values generated by probability distributions, values that assume some importance in studies of, for instance, earthquakes, other natural disasters, and the risks thereof; see [14].

of the scaling parameter, the fit to the remaining data has a simple functional form that allows us to easily test the level of agreement between the data and the best-fit model, as discussed in section 5.

4. Testing the Power-Law Hypothesis. The tools described in the previous sections allow us to fit a power-law distribution to a given data set and provide estimates of the parameters α and $x_{\min}$. They tell us nothing, however, about whether the power law is a *plausible* fit to the data. Regardless of the true distribution from which our data were drawn, we can always fit a power law. We need some way to tell whether the fit is a good match to the data.

Most previous empirical studies of ostensibly power-law distributed data have not attempted to test the power-law hypothesis quantitatively. Instead, they typically rely on qualitative appraisals of the data, based, for instance, on visualizations. But these can be deceptive and can lead to claims of power-law behavior that do not hold up under closer scrutiny. Consider Figure 5a, which shows the CDFs of three small data sets ($n = 100$) drawn from a power-law distribution with $\alpha = 2.5$, a log-normal distribution with $\mu = 0.3$ and $\sigma = 2.0$, and an exponential distribution with exponential parameter $\lambda = 0.125$. In each case the distributions have a lower bound of $x_{\min} = 15$. Because each of these distributions looks roughly straight on the log-log plot used in the figure, one might, upon cursory inspection, judge all three to follow power laws, albeit with different scaling parameters. This judgment would, however, be wrong—being roughly straight on a log-log plot is a necessary but not sufficient condition for power-law behavior.

Unfortunately, it is not straightforward to say with certainty whether a particular data set has a power-law distribution. Even if data are drawn from a power law their observed distribution is extremely unlikely to exactly follow the power-law form; there will always be some small deviations because of the random nature of the sampling process. The challenge is to distinguish deviations of this type from those that arise because the data are drawn from a non-power-law distribution.

The basic approach, as we describe in this section, is to sample many synthetic data sets from a true power-law distribution, measure how far they fluctuate from the power-law form, and compare the results with similar measurements on the empirical data. If the empirical data set is much further from the power-law form than the typical synthetic one, then the power law is not a plausible fit to the data. Two notes of caution are worth sounding. First, the effectiveness of this approach depends on how we measure the distance between distributions. Here, we use the KS statistic, which typically gives good results, but in principle another goodness-of-fit measure could be used in its place. Second, it is of course always possible that a non-power-law process will, as a result again of sampling fluctuations, happen to generate a data set with a distribution close to a power law, in which case our test will fail. The odds of this happening, however, dwindle with increasing n , which is the primary reason why one prefers large statistical samples when attempting to verify hypotheses such as these.

4.1. Goodness-of-Fit Tests. Given an observed data set and a hypothesized power-law distribution from which the data are drawn, we would like to know whether our hypothesis is a plausible one, given the data.

A standard approach to answering this kind of question is to use a *goodness-of-fit test*, which generates a p -value that quantifies the plausibility of the hypothesis. Such tests are based on measurement of the “distance” between the distri-

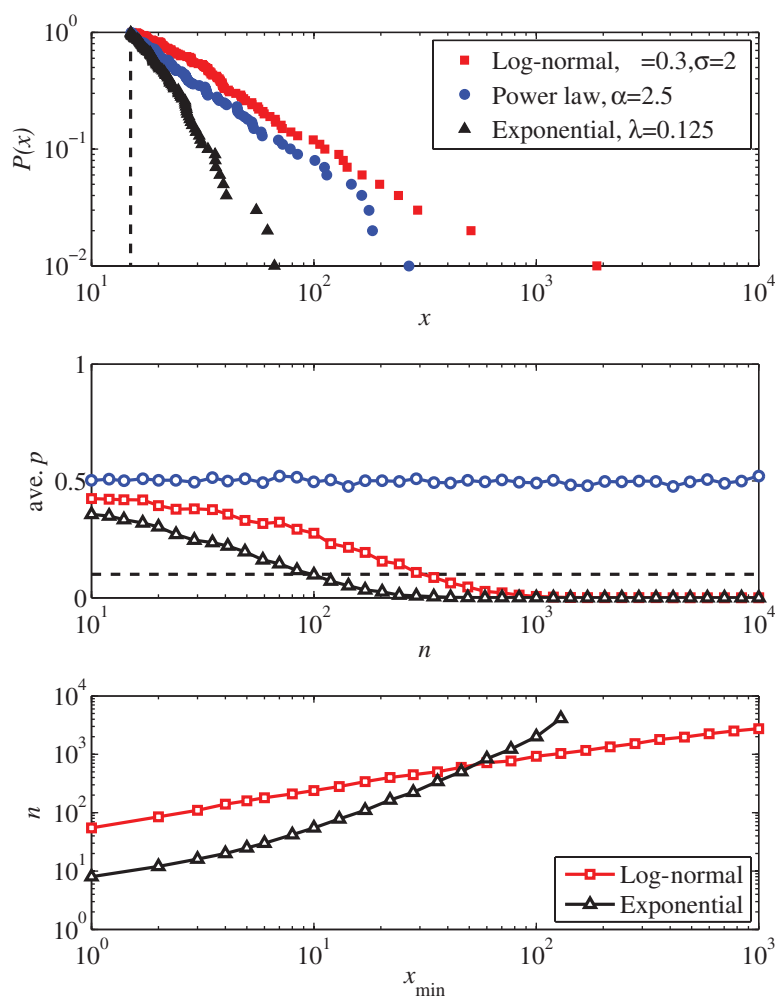


Fig. 5 (a) The CDFs of three small samples ($n = 100$) drawn from different continuous distributions: a log-normal with $\mu = 0.3$ and $\sigma = 2$, a power law with $\alpha = 2.5$, and an exponential with $\lambda = 0.125$, all with $x_{\min} = 15$. (Definitions of the parameters are as in Table 1.) Visually, each of the CDFs appears roughly straight on the logarithmic scales used, but only one is a true power law. (b) The average p -value for the maximum likelihood power-law model for samples from the same three distributions, as a function of the number of observations n . As n increases, only the p -value for power-law distributed data remains above our rule-of-thumb threshold $p = 0.1$, with the others falling off toward zero, indicating that p does correctly identify the true power-law behavior in this case. (c) The average number of observations n required to reject the power-law hypothesis (i.e., to make $p < 0.1$) for data drawn from the log-normal and exponential distributions, as a function of $x_{\min}$.

bution of the empirical data and the hypothesized model. This distance is compared with distance measurements for comparable synthetic data sets drawn from the same model, and the p -value is defined to be the fraction of the synthetic distances that are larger than the empirical distance. If p is large (close to 1), then the difference between the empirical data and the model can be attributed to statistical fluctuations alone; if it is small, the model is not a plausible fit to the data.

As we have seen in sections 3.3 and 3.4 there are a variety of measures for quantifying the distance between two distributions. In our calculations we use the KS statistic, which we encountered in section 3.3.⁷ In detail, our procedure is as follows.

First, we fit our empirical data to the power-law model using the methods of section 3 and calculate the KS statistic for this fit. Next, we generate a large number of power-law distributed synthetic data sets with scaling parameter α and lower bound $x_{\min}$ equal to those of the distribution that best fits the observed data. We fit each synthetic data set individually to its own power-law model and calculate the KS statistic for each one relative to its own model. Then we simply count the fraction of the time that the resulting statistic is larger than the value for the empirical data. This fraction is our p -value.

Note that for each synthetic data set we compute the KS statistic relative to the best-fit power law for that data set, not relative to the original distribution from which the data set was drawn. In this way we ensure that we are performing for each synthetic data set the same calculation that we performed for the real data set, a crucial requirement if we wish to get an unbiased estimate of the p -value.

The generation of the synthetic data involves some subtleties. To obtain accurate estimates of p we need synthetic data that have a distribution similar to the empirical data below $x_{\min}$ but that follow the fitted power law above $x_{\min}$. To generate such data we make use of a semiparametric approach. Suppose that our observed data set has n_{tail} observations $x \geq x_{\min}$ and n observations in total. We generate a new data set with n observations as follows. With probability n_{tail}/n we generate a random number x_i drawn from a power law with scaling parameter $\hat{\alpha}$ and $x \geq x_{\min}$. Otherwise, with probability $1 - n_{\text{tail}}/n$, we select one element uniformly at random from among the elements of the observed data set that have $x < x_{\min}$ and set x_i equal to that element. Repeating the process for all $i = 1, \dots, n$ we generate a complete synthetic data set that indeed follows a power law above $x_{\min}$ but has the same (non-power-law) distribution as the observed data below.

We also need to decide how many synthetic data sets to generate. Based on an analysis of the expected worst-case performance of the test, a good rule of thumb turns out to be the following: if we wish our p -values to be accurate to within about ϵ of the true value, then we should generate at least $\frac{1}{4}\epsilon^{-2}$ synthetic data sets. Thus, if we wish our p -value to be accurate to about 2 decimal digits, we should choose $\epsilon = 0.01$, which implies we should generate about 2500 synthetic sets. For the example calculations described in section 6 we used numbers of this order, ranging from 1000 to 10 000 depending on the particular application.

Once we have calculated our p -value, we need to make a decision about whether it is small enough to rule out the power-law hypothesis or whether, conversely, the hypothesis is a plausible one for the data in question. In our calculations we have made the relatively conservative choice that the power law is ruled out if $p \leq 0.1$; that is, it is ruled out if there is a probability of 1 in 10 or less that we would merely by chance get data that agree as poorly with the model as the data we have. (In other contexts, many authors use the more lenient rule $p \leq 0.05$, but we feel this would let through some candidate distributions that have only a very small chance of

⁷One of the nice features of the KS statistic is that its distribution is known for data sets truly drawn from any given distribution. This allows one to write down an explicit expression in the limit of large n for the p -value; see, for example, [46]. Unfortunately, this expression is only correct so long as the underlying distribution is fixed. If, as in our case, the underlying distribution is itself determined by fitting to the data and hence varies from one data set to the next, we cannot use this approach, which is why we recommend the Monte Carlo procedure described here instead.

really following a power law. Of course, in practice, the particular rule adopted must depend on the judgment of the investigator and the circumstances at hand.⁸)

It is important to appreciate that a large p -value does not necessarily mean the power law is the correct distribution for the data. There are (at least) two reasons for this. First, there may be other distributions that match the data equally well or better over the range of x observed. Other tests are needed to rule out such alternatives, which we discuss in section 5.

Second, as mentioned above, it is possible for small values of n that the empirical distribution will follow a power law closely, and hence that the p -value will be large, even when the power law is the wrong model for the data. This is not a deficiency of the method; it reflects the fact that it is genuinely harder to rule out the power law if we have very little data. For this reason, high p -values should be treated with caution when n is small.

4.2. Performance of the Goodness-of-Fit Test. To demonstrate the utility of this approach, and to show that it can correctly distinguish power-law from non-power-law behavior, we consider data of the type shown in Figure 5a, drawn from continuous power-law, log-normal, and exponential distributions. In Figure 5b we show the average p -value, calculated as above, for data sets drawn from these three distributions, as a function of the number of samples n . When n is small, meaning $n \lesssim 100$ in this case, the p -values for all three distributions are above our threshold of 0.1, meaning that the power-law hypothesis is not ruled out by our test—for samples this small we cannot accurately distinguish the data sets because there is simply not enough data to go on. As the sizes of the samples become larger, however, the p -values for the two non-power-law distributions fall off and it becomes possible to say that the power-law model is a poor fit for these data sets, while remaining a good fit for the true power-law data set.

It is important to note, however, that, since we fit the power-law form to only the part of the distribution above $x_{\min}$, the value of $x_{\min}$ effectively controls the number of data points we have to work with. If $x_{\min}$ is large, then only a small fraction of the data set falls above it and thus the larger the value of $x_{\min}$, the larger the total value of n needed to reject the power law. This phenomenon is depicted in Figure 5c, which shows the value of n needed to cross below the threshold value of $p = 0.1$ for the log-normal and exponential distributions as a function of $x_{\min}$.

5. Alternative Distributions. The method described in section 4 provides a reliable way to test whether a given data set is plausibly drawn from a power-law distribution. However, the results of such tests don't tell the whole story. Even if our data are well fit by a power law, it is still possible that another distribution, such as an exponential or a log-normal, might give a fit as good or better. We can eliminate this possibility by using a goodness-of-fit test again—we can simply calculate a p -value for a fit to the competing distribution and compare it to the p -value for the power law.

Suppose, for instance, that we believe our data might follow either a power-law or an exponential distribution. If we discover that the p -value for the power law is reasonably large (say, $p > 0.1$), then the power law is not ruled out. To strengthen

⁸Some readers will be familiar with the use of p -values to confirm (rather than rule out) hypotheses for experimental data. In the latter context, one quotes a p -value for a “null” model, a model *other* than the model the experiment is attempting to verify. Normally one then considers low values of p to be good, since they indicate that the null hypothesis is unlikely to be correct. Here, by contrast, we use the p -value as a measure of the hypothesis we are trying to verify, and hence high values, not low, are “good.” For a general discussion of the interpretation of p -values, see [39].

our case for the power law we would like to rule out the competing exponential distribution, if possible. To do this, we would find the best-fit exponential distribution, using the equivalent for exponentials of the methods of section 3, and the corresponding KS statistic, then repeat the calculation for a large number of synthetic data sets and hence calculate a p -value. If the p -value is sufficiently small, we can rule out the exponential as a model for our data.

By combining p -value calculations with respect to the power law and several plausible competing distributions, we can in this way make a good case for or against the power-law form for our data. In particular, if the p -value for the power law is high, while those for competing distributions are small, then the competition is ruled out and, although we cannot say absolutely that the power law is correct, the case in its favor is strengthened.

We cannot of course compare the power-law fit of our data with fits to every competing distribution, of which there is an infinite number. Indeed, as is usually the case with data fitting, it will almost always be possible to find a class of distributions that fits the data better than the power law if we define a family of curves with a sufficiently large number of parameters. Fitting the statistical distribution of data should therefore be approached using a combination of statistical techniques like those described here and prior knowledge about what constitutes a reasonable model for the data. Statistical tests can be used to rule out specific hypotheses, but it is up to the researcher to decide what a reasonable hypothesis is in the first place.

5.1. Direct Comparison of Models. The methods of the previous section can tell us whether either or both of two candidate distributions—usually the power-law distribution and some alternative—can be ruled out as a fit to our data or, if neither is ruled out, which is the better fit. In many practical situations, however, we only want to know the latter—which distribution is the better fit. This is because we will normally have already performed a goodness-of-fit test for the first distribution, the power law. If that test fails and the power law is rejected, then our work is done and we can move on to other things. If it passes, on the other hand, then our principal concern is whether another distribution might provide a better fit.

In such cases, methods exist which can directly compare two distributions against each other and which are considerably easier to implement than the KS test. In this section we describe one such method, the *likelihood ratio test*.⁹

The basic idea behind the likelihood ratio test is to compute the likelihood of the data under two competing distributions. The one with the higher likelihood is then the better fit. Alternatively, one can calculate the ratio of the two likelihoods, or equivalently the logarithm $\mathcal{R}$ of the ratio, which is positive or negative depending on which distribution is better, or zero in the event of a tie.

The sign of the log-likelihood ratio alone, however, will not definitively indicate which model is the better fit because, like other quantities, it is subject to statistical fluctuation. If its true value, meaning its expected value over many independent data sets drawn from the same distribution, is close to zero, then the fluctuations could change the sign of the ratio and hence the results of the test cannot be trusted. In order to make a firm choice between distributions we need a log-likelihood ratio that is sufficiently positive or negative that it could not plausibly be the result of a chance fluctuation from a true result that is close to zero.

⁹The likelihood ratio test is not the only possible approach. Others include fully Bayesian approaches [31], cross-validation [58], or minimum description length (MDL) [20].

To make a quantitative judgment about whether the observed value of $\mathcal{R}$ is sufficiently far from zero, we need to know the size of the expected fluctuations; that is, we need to know the standard deviation σ on $\mathcal{R}$. This we can estimate from our data using a method proposed by Vuong [62]. This method gives a p -value that tells us whether the observed sign of $\mathcal{R}$ is statistically significant. If this p -value is small (say, $p < 0.1$), then it is unlikely that the observed sign is a chance result of fluctuations and the sign is a reliable indicator of which model is the better fit to the data. If p is large, on the other hand, the sign is not reliable and the test does not favor either model over the other. It is one of the advantages of this approach that it can tell us not only which of two hypotheses is favored, but also when the data are insufficient to favor either of them.¹⁰ The simple goodness-of-fit test of the previous section provides no equivalent indication when the data are insufficient.¹¹ The technical details of the likelihood ratio test are described in Appendix C.

5.2. Nested Hypotheses. In some cases the distributions we wish to compare may be *nested*, meaning that one family of distributions is a subset of the other. The power law and the power law with exponential cutoff in Table 1 provide an example of such nested distributions. When distributions are nested it is always the case that the larger family of distributions will provide a fit at least as good as the smaller, since every member of the smaller family is also a member of the larger. In this case, a slightly modified likelihood ratio test is needed to properly distinguish between such models, as described in Appendix C.

5.3. Performance of the Likelihood Ratio Test. As with the other methods discussed here, we can quantify the performance of the likelihood ratio test by applying it to synthetic data. For our tests, we generated data from two distributions: a continuous power law with $\alpha = 2.5$ and $x_{\min} = 1$, and a log-normal distribution with $\mu = 0.3$ and $\sigma = 2$ constrained to only produce positive values of x . (These are the same parameter values we used in section 4.2.) In each case we drew n independent values from each distribution and estimated the value of $x_{\min}$ for each set of values, then calculated the likelihood ratio for the data above $x_{\min}$ and the corresponding p -value. This procedure was repeated 1000 times to assess sampling fluctuations. Following Vuong [62] we calculated the normalized log-likelihood ratio $n^{-1/2}\mathcal{R}/\sigma$, where σ is the estimated standard deviation on $\mathcal{R}$. The normalized figure is in many ways more convenient than the raw one since the p -value can be calculated directly from it using eq. (C.6). (In a sense this makes it unnecessary to actually calculate p since the normalized log-likelihood ratio contains the same information, but it is convenient when making judgments about particular cases to have the actual p -value at hand, so we give both in our results.)

¹⁰In cases where we are unable to distinguish between two hypothesized distributions, one could claim that there is really no difference between them: if both are good fits to the data, then it makes no difference which one we use. This may be true in some cases, but it is certainly not true in general. In particular, if we wish to extrapolate a fitted distribution far into its tail, to predict, for example, the frequencies of large but rare events like major earthquakes or meteor impacts, then conclusions based on different fitted forms can differ enormously even if the forms are indistinguishable in the domain covered by the actual data. Thus the ability to say whether the data clearly favor one hypothesis over another can have substantial practical consequences.

¹¹One alternative method for choosing between distributions, the Bayesian approach described in [59], is essentially equivalent to the likelihood ratio test, but without the p -value to tell us when the results are significant. The Bayesian estimation used is equivalent to a smoothing, which to some extent buffers the results against the effects of fluctuations [52], but the method itself is not capable of determining whether the results could be due to chance [38, 64].

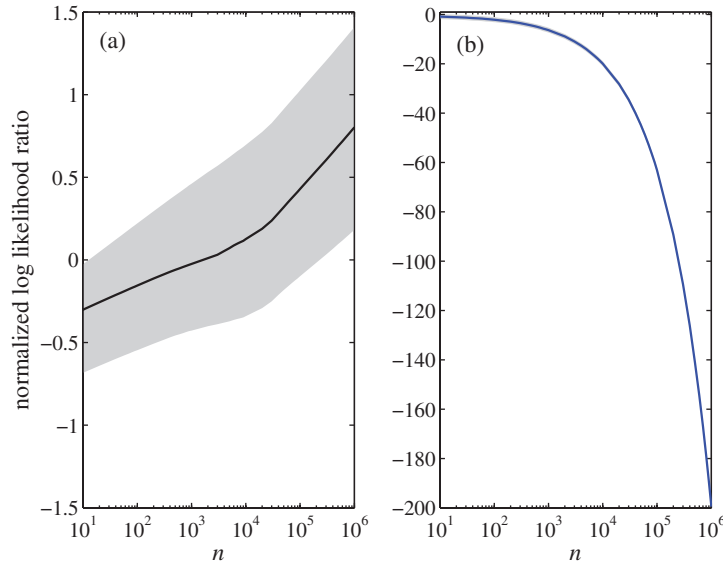


Fig. 6 Behavior of the normalized log-likelihood ratio $n^{-1/2}\mathcal{R}/\sigma$ for synthetic data sets of n points drawn from either (a) a continuous power law with $\alpha = 2.5$ and $x_{\min} = 1$ or (b) a log-normal with $\mu = 0.3$ and $\sigma = 2$. Results are averaged over 1000 replications at each sample size, and the range covered by the 1st to 3rd quartiles is shown in gray.

Figure 6 shows the behavior of the normalized log-likelihood ratio as a function of n . As the figure shows, it becomes increasingly positive as n grows for data drawn from a true power law, but increasingly negative for data drawn from a log-normal.

If we ignore the p -value and simply classify each of our synthetic data sets as power-law or log-normal according to the raw sign of the log-likelihood ratio $\mathcal{R}$, then, as we have said, we will sometimes reach the wrong conclusion if $\mathcal{R}$ is close to zero and we are unlucky with the sampling fluctuations. Figure 7a shows the fraction of data sets misclassified in this way in our tests as a function of n , and though the numbers decrease with sample size n , they are uncomfortably large for moderate values. If we take the p -value into account, however, using its value to perform a more nuanced classification as power-law, log-normal, or undecided, as described above, the fraction of misclassifications is far better, falling to a few parts per thousand, even for quite modest sample sizes—see Figure 7b. These results indicate that the p -value is effective at identifying cases in which the data are insufficient to make a firm distinction between hypotheses.

6. Applications to Real-World Data. In this section, as a demonstration of the utility of the methods described in this paper, we apply them to a variety of real-world data sets representing measurements of quantities whose distributions have been conjectured to follow power laws. As we will see, the results indicate that some of the data sets are indeed consistent with a power-law hypothesis, but others are not, and some are marginal cases for which the power law is a possible candidate distribution, but is not strongly supported by the data.

The 24 data sets we study are drawn from a broad variety of different branches of human endeavor, including physics, earth sciences, biology, ecology, paleontology, computer and information sciences, engineering, and the social sciences. They are as follows:

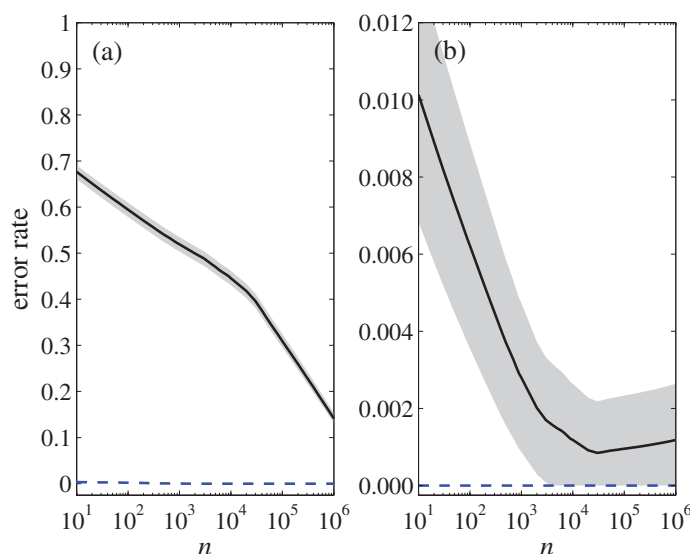


Fig. 7 Rates of misclassification of distributions by the likelihood ratio test if (a) the p -value is ignored and classification is based only on the sign of the log-likelihood ratio, and (b) if the p -value is taken into account and we count only misclassifications where the log-likelihood ratio has the wrong sign and the p -value is less than 0.05. Results are for the same synthetic data as Figure 6. The black line shows the rate of misclassification (over 1000 repetitions) of power-law samples as log-normals (95% confidence interval shown in gray), while the (dashed) line shows the rate of misclassification of log-normals as power laws (95% confidence interval is smaller than the width of the line).

- (a) The frequency of occurrence of unique words in the novel *Moby Dick* by Herman Melville [43].
- (b) The degrees (i.e., numbers of distinct interaction partners) of proteins in the partially known protein-interaction network of the yeast *Saccharomyces cerevisiae* [28].
- (c) The degrees of metabolites in the metabolic network of the bacterium *Escherichia coli* [26].
- (d) The degrees of nodes in the partially known network representation of the Internet at the level of autonomous systems for May 2006 [25]. (An autonomous system is a group of IP addresses on the Internet among which routing is handled internally or “autonomously,” rather than using the Internet’s large-scale border gateway protocol routing mechanism.)
- (e) The number of calls received by customers of AT&T’s long distance telephone service in the United States during a single day [1, 5].
- (f) The intensity of wars from 1816–1980 measured as the number of battle deaths per 10 000 of the combined populations of the warring nations [53, 49].
- (g) The severity of terrorist attacks worldwide from February 1968 to June 2006, measured as the number of deaths directly resulting [11].
- (h) The number of bytes of data received as the result of individual web (HTTP) requests from computer users at a large research laboratory during a 24-hour period in June 1996 [68]. Roughly speaking, this distribution represents the size distribution of web files transmitted over the Internet.
- (i) The number of species per genus of mammals. This data set, compiled by Smith et al. [54], is composed primarily of species alive today but also includes

some recently extinct species, where “recent” in this context means the last few tens of thousands of years.

- (j) The numbers of sightings of birds of different species in the North American Breeding Bird Survey for 2003.
- (k) The numbers of customers affected in electrical blackouts in the United States between 1984 and 2002 [43].
- (l) The numbers of copies of bestselling books sold in the United States during the period 1895 to 1965 [21].
- (m) The human populations of U.S. cities in the 2000 U.S. Census.
- (n) The sizes of email address books of computer users at a large university [44].
- (o) The sizes in acres of wildfires occurring on U.S. federal land between 1986 and 1996 [43].
- (p) Peak gamma-ray intensity of solar flares between 1980 and 1989 [43].
- (q) The intensities of earthquakes occurring in California between 1910 and 1992, measured as the maximum amplitude of motion during the quake [43].
- (r) The numbers of adherents of religious denominations, bodies, and sects, as compiled and published on the web site adherents.com.
- (s) The frequencies of occurrence of U.S. family names in the 1990 U.S. Census.
- (t) The aggregate net worth in U.S. dollars of the richest individuals in the United States in October 2003 [43].
- (u) The number of citations received between publication and June 1997 by scientific papers published in 1981 and listed in the Science Citation Index [47].
- (v) The number of academic papers authored or coauthored by mathematicians listed in the American Mathematical Society’s MathSciNet database. (Data compiled by J. Grossman.)
- (w) The number of “hits” received by web sites from customers of the America Online Internet service in a single day [3].
- (x) The number of links to web sites found in a 1997 web crawl of about 200 million web pages [10].

Many of these data sets are only subsets of much larger entities (such as the web sites, which are only a small fraction of the entire web). In some cases it is known that the sampling procedure used to obtain these subsets may be biased, as, for example, in the protein interactions [56], citations and authorships [9], and the Internet [2, 15]. We have not attempted to correct any biases in our analysis.

In Table 3 we show results from the fitting of a power-law form to each of these data sets using the methods described in section 3, along with a variety of generic statistics for the data such as mean, standard deviation, and maximum value. In the last column of the table we give the p -value for the power-law model, estimated as in section 4, which gives a measure of how plausible the power law is as a fit to the data. Figures 8 and 9 show these data graphically, along with the estimated power-law distributions.

As an indication of the importance of accurate methods for fitting power-law data, we note that many of our values for the scaling parameters differ considerably from those derived from the same data by previous authors using ad hoc methods. For instance, the scaling parameter for the protein interaction network of [28] has been reported to take a value of 2.44 [69], which is quite different from, and incompatible with, the value we find of 3.1 ± 0.3 . Similarly, the citation distribution data of [47] have been reported to have a scaling parameter of either 2.9 [61] or 2.5 [32], neither of which are compatible with our maximum likelihood figure of 3.16 ± 0.06 .

Table 3 Basic parameters of the data sets described in section 6, along with their power-law fits and the corresponding p -values (statistically significant values are denoted in **bold**).

Quantity	n	$\langle x \rangle$	σ	$x_{\max}$	$\hat{x}_{\min}$	$\hat{\alpha}$	n_{tail}	p
count of word use	18 855	11.14	148.33	14 086	7 ± 2	$1.95(2)$	2958 ± 987	0.49
protein interaction degree	1846	2.34	3.05	56	5 ± 2	$3.1(3)$	204 ± 263	0.31
metabolic degree	1641	5.68	17.81	468	4 ± 1	$2.8(1)$	748 ± 136	0.00
Internet degree	22 688	5.63	37.83	2583	21 ± 9	$2.12(9)$	770 ± 1124	0.29
telephone calls received	51 360 423	3.88	179.09	375 746	120 ± 49	$2.09(1)$	$102 592 \pm 210 147$	0.63
intensity of wars	115	15.70	49.97	382	2.1 ± 3.5	$1.7(2)$	70 ± 14	0.20
terrorist attack severity	9101	4.35	31.58	2749	12 ± 4	$2.4(2)$	547 ± 1663	0.68
HTTP size (kilobytes)	226 386	7.36	57.94	10 971	36.25 ± 22.74	$2.48(5)$	6794 ± 2232	0.00
species per genus	509	5.59	6.94	56	4 ± 2	$2.4(2)$	233 ± 138	0.10
bird species sightings	591	3384.36	10 952.34	138 705	6679 ± 2463	$2.1(2)$	66 ± 41	0.55
blackouts ($\times 10^3$)	211	253.87	610.31	7500	230 ± 90	$2.3(3)$	59 ± 35	0.62
sales of books ($\times 10^3$)	633	1986.67	1396.60	19 077	2400 ± 430	$3.7(3)$	139 ± 115	0.66
population of cities ($\times 10^3$)	19 447	9.00	77.83	8 009	52.46 ± 11.88	$2.37(8)$	580 ± 177	0.76
email address books size	4581	12.45	21.49	333	57 ± 21	$3.5(6)$	196 ± 449	0.16
forest fire size (acres)	203 785	0.90	20.99	4121	6324 ± 3487	$2.2(3)$	521 ± 6801	0.05
solar flare intensity	12 773	689.41	6520.59	231 300	323 ± 89	$1.79(2)$	1711 ± 384	1.00
quake intensity ($\times 10^3$)	19 302	24.54	563.83	63 096	0.794 ± 80.198	$1.64(4)$	$11 697 \pm 2159$	0.00
religious followers ($\times 10^6$)	103	27.36	136.64	1050	3.85 ± 1.60	$1.8(1)$	39 ± 26	0.42
freq. of surnames ($\times 10^3$)	2753	50.59	113.99	2502	111.92 ± 40.67	$2.5(2)$	239 ± 215	0.20
net worth (mil. USD)	400	2388.69	4 167.35	46 000	900 ± 364	$2.3(1)$	302 ± 77	0.00
citations to papers	415 229	16.17	44.02	8904	160 ± 35	$3.16(6)$	3455 ± 1859	0.20
papers authored	401 445	7.21	16.52	1416	133 ± 13	$4.3(1)$	988 ± 377	0.90
hits to web sites	119 724	9.83	392.52	129 641	2 ± 13	$1.81(8)$	$50 981 \pm 16 898$	0.00
links to web sites	241 428 853	9.15	106 871.65	1 199 466	3684 ± 151	$2.336(9)$	$28 986 \pm 1560$	0.00

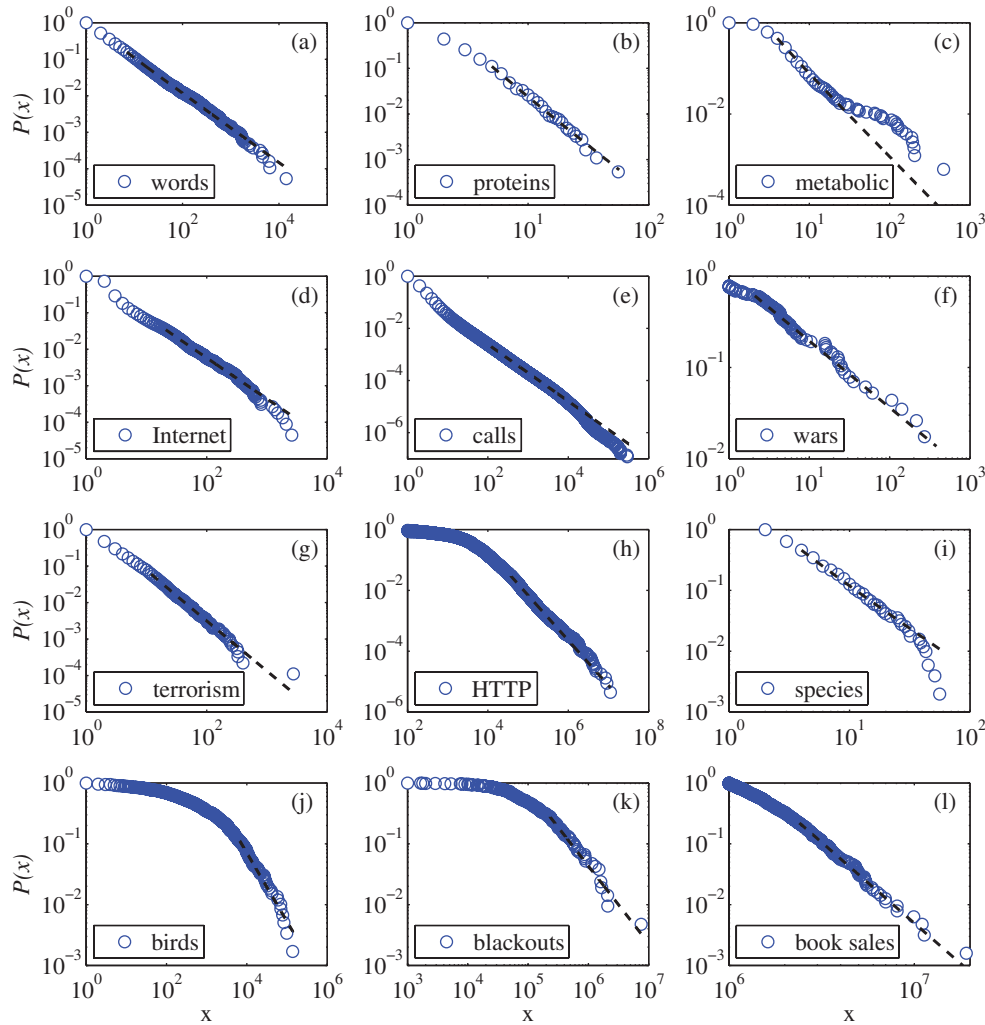


Fig. 8 The CDFs $P(x)$ and their maximum likelihood power-law fits for the first 12 of our 24 empirical data sets. (a) The frequency of occurrence of unique words in the novel *Moby Dick* by Herman Melville. (b) The degree distribution of proteins in the protein interaction network of the yeast *S. cerevisiae*. (c) The degree distribution of metabolites in the metabolic network of the bacterium *E. coli*. (d) The degree distribution of autonomous systems (groups of computers under single administrative control) on the Internet. (e) The number of calls received by U.S. customers of the long-distance telephone carrier AT&T. (f) The intensity of wars from 1816–1980 measured as the number of battle deaths per 10 000 of the combined populations of the warring nations. (g) The severity of terrorist attacks worldwide from February 1968 to June 2006, measured by number of deaths. (h) The number of bytes of data received in response to HTTP (web) requests from computers at a large research laboratory. (i) The number of species per genus of mammals during the late Quaternary period. (j) The frequency of sightings of bird species in the United States. (k) The number of customers affected by electrical blackouts in the United States. (l) The sales volume of bestselling books in the United States.

The p -values in Table 3 indicate that 17 of the 24 data sets are consistent with a power-law distribution. The remaining seven data sets all have p -values small enough that the power-law model can be firmly ruled out. In particular, the distributions for the HTTP connections, earthquakes, web links, fires, wealth, web hits, and the

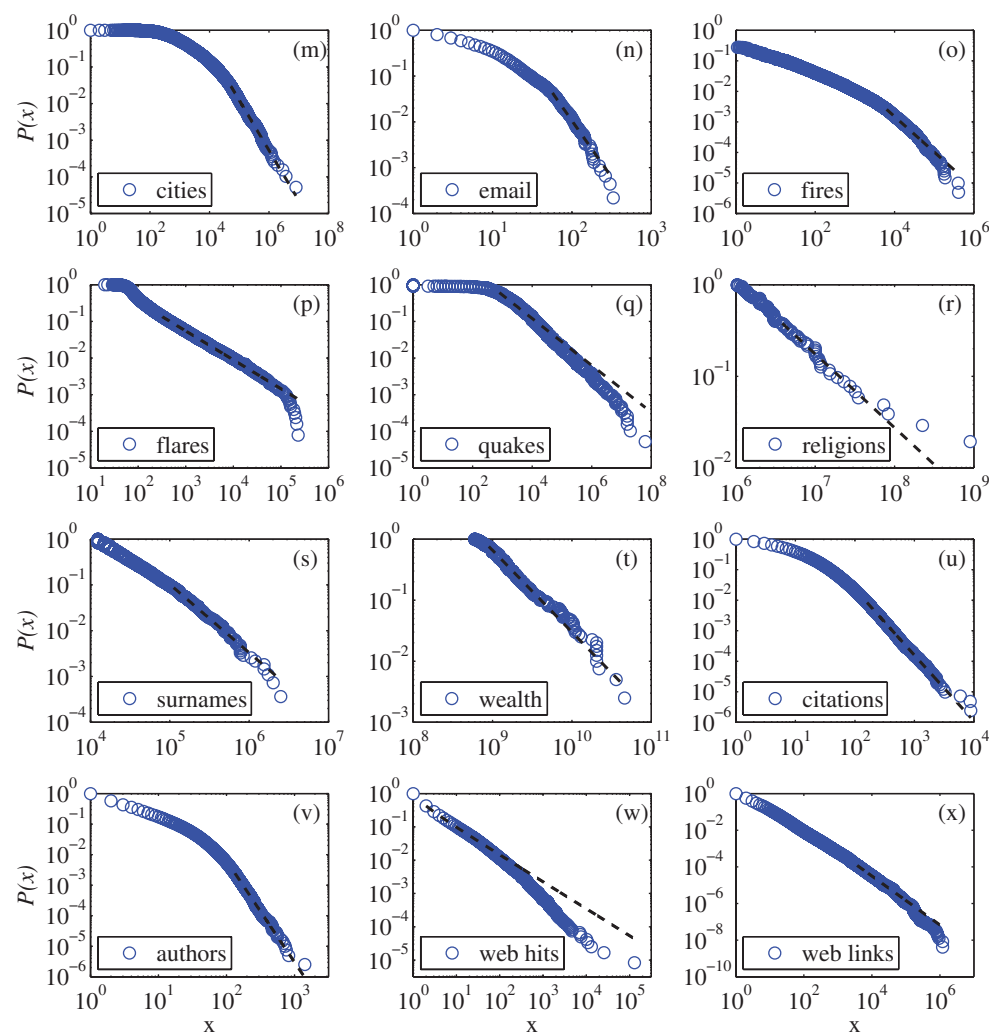


Fig. 9 The CDFs $P(x)$ and their maximum likelihood power-law fits for the second 12 of our 24 empirical data sets. (m) The populations of cities in the United States. (n) The sizes of email address books at a university. (o) The number of acres burned in California forest fires. (p) The intensities of solar flares. (q) The intensities of earthquakes. (r) The numbers of adherents of religious sects. (s) The frequencies of surnames in the United States. (t) The net worth in U.S. dollars of the richest people in the United States. (u) The numbers of citations received by published academic papers. (v) The numbers of papers authored by mathematicians. (w) The numbers of hits on web sites from AOL users. (x) The numbers of hyperlinks to web sites.

metabolic network cannot plausibly be considered to follow a power law; the probability of getting by chance a fit as poor as the one observed is very small in each of these cases and one would have to be unreasonably optimistic to see power-law behavior in any of these data sets. (For two data sets—the HTTP connections and wealth distribution—the power law, while not a good fit, is nonetheless better than the alternatives we tested using the likelihood ratio test, implying that these data sets are not well characterized by any of the functional forms considered here.)

Tables 4 and 5 show the results of likelihood ratio tests comparing the best-fit power laws for each of our data sets to the alternative distributions given in Table 1.

Table 4 Tests of power-law behavior in the data sets with continuous (real) data. (Results for the discrete data sets are given in Table 5.) For each data set we give a p -value for the fit to the power-law model and likelihood ratios for the alternatives. We also quote p -values for the significance of each of the likelihood ratio tests. Statistically significant p -values are denoted in **bold**. Positive values of the log-likelihood ratios indicate that the power-law model is favored over the alternative. For nonnested alternatives, we give the normalized log-likelihood ratio $n^{-1/2}\mathcal{R}/\sigma$ which appears in eq. (C.6), while for the power law with exponential cut-off we give the actual log-likelihood ratio. The final column of the table lists our judgment of the statistical support for the power-law hypothesis for each data set. “None” indicates data sets that are probably not power-law distributed; “moderate” indicates that the power law is a good fit but that there are other plausible alternatives as well; “good” indicates that the power law is a good fit and that none of the alternatives considered is plausible. (None of the data sets in this table earned a rating of “good,” but one data set in Table 5, for the frequencies of words, is so designated.) In some cases, we write “with cut-off,” meaning that the power law with exponential cutoff is clearly favored over the pure power law. In each of the latter cases, however, some of the alternative distributions are also good fits, such as the log-normal or the stretched exponential distribution.

Data set	Power law		Log-normal		Exponential		Stretched exp.		Power law + cut-off		Support for power law
	p	LR	p	LR	p	LR	p	LR	p		
birds	0.55	-0.850	0.40	1.87	0.06	-0.882	0.38	-1.24	0.12		moderate
blackouts	0.62	-0.412	0.68	1.21	0.23	-0.417	0.68	-0.382	0.38		moderate
book sales	0.66	-0.267	0.79	2.70	0.01	3.885	0.00	-0.140	0.60		moderate
cities	0.76	-0.090	0.93	3.65	0.00	0.204	0.84	-0.123	0.62		moderate
fires	0.05	-1.78	0.08	4.00	0.00	-1.82	0.07	-5.02	0.00		with cut-off
flares	1.00	-0.803	0.42	13.7	0.00	-0.546	0.59	-4.52	0.00		with cut-off
HTTP	0.00	1.77	0.08	11.8	0.00	2.65	0.01	0.000	1.00		none
quakes	0.00	-7.14	0.00	11.6	0.00	-7.09	0.00	-24.4	0.00		with cut-off
religions	0.42	-0.073	0.94	1.59	0.11	1.75	0.08	-0.167	0.56		moderate
surnames	0.20	-0.836	0.40	2.89	0.00	-0.844	0.40	-1.36	0.10		with cut-off
wars	0.20	-0.737	0.46	3.68	0.00	-0.767	0.44	-0.847	0.19		moderate
wealth	0.00	0.249	0.80	6.20	0.00	8.05	0.00	-0.142	0.59		none
web hits	0.00	-10.21	0.00	8.55	0.00	10.94	0.00	-74.66	0.00		with cut-off
web links	0.00	-2.24	0.03	25.3	0.00	-1.08	0.28	-21.2	0.00		with cut-off

Table 5 Tests of power-law behavior in the data sets with discrete (integer) data. Statistically significant p -values are denoted in **bold**. Results for the continuous data sets are given in Table 4; see that table for a description of the individual column entries.

Data set	p	Poisson		Log-normal		Exponential		Stretched exp.		Power law + cut-off		Support for power law with cut-off
		LR	p	LR	p	LR	p	LR	p	LR	p	
Internet	0.29	5.31	0.00	-0.807	0.42	6.49	0.00	0.493	0.62	-1.97	0.05	power law
calls	0.63	17.9	0.00	-2.03	0.04	35.0	0.00	14.3	0.00	-30.2	0.00	with cut-off
citations	0.20	6.54	0.00	-0.141	0.89	5.91	0.00	1.72	0.09	-0.007	0.91	moderate
email	0.16	4.65	0.00	-1.10	0.27	0.639	0.52	-1.13	0.26	-1.89	0.05	with cut-off
metabolic	0.00	3.53	0.00	-1.05	0.29	5.59	0.00	3.66	0.00	0.000	1.00	none
papers	0.90	5.71	0.00	-0.091	0.93	3.08	0.00	0.709	0.48	-0.016	0.86	moderate
proteins	0.31	3.05	0.00	-0.456	0.65	2.21	0.03	0.055	0.96	-0.414	0.36	moderate
species	0.10	5.04	0.00	-1.63	0.10	2.39	0.02	-1.59	0.11	-3.80	0.01	with cut-off
terrorism	0.68	1.81	0.07	-0.278	0.78	2.457	0.01	0.772	0.44	-0.077	0.70	moderate
words	0.49	4.43	0.00	0.395	0.69	9.09	0.00	4.13	0.00	-0.899	0.18	good

For reference, the first column repeats the p -values given in Table 3. Based on the results of our tests, we summarize in the final column of the table how convincing the power-law model is as a fit to each data set.

There is only one case—the distribution of the frequencies of occurrence of words in English text—in which the power law appears to be truly convincing, in the sense that it is an excellent fit to the data and none of the alternatives carries any weight.

Among the remaining data sets we can rule out the exponential distribution as a possible fit in all cases save three. The three exceptions are the blackouts, religions, and email address books, for which the power law is favored over the exponential but the accompanying p -value is large enough that the results cannot be trusted. For the discrete data sets (Table 5) we can also rule out the Poisson distribution in every case.

The results for the log-normal and stretched exponential distributions are more ambiguous; in most cases the p -values for the log-likelihood ratio tests are sufficiently large that the results of the tests are inconclusive. In particular, the distributions for birds, books, cities, religions, wars, citations, papers, proteins, and terrorism are plausible power laws, but they are also plausible log-normals and stretched exponentials. In cases such as these, it is important to look at physical motivating or theoretical factors to make a sensible judgment about which distributional form is more reasonable—we must consider whether there is a mechanistic or other non-statistical argument favoring one distribution or another. The specific problem of the indistinguishability of power laws and stretched exponentials has also been discussed by Malevergne, Pisarenko, and Sornette [35].

In some other cases the likelihood ratio tests do give conclusive answers. For instance, the stretched exponential is ruled out for the book sales, telephone calls, and citation counts, but is strongly favored over the power law for the forest fires and earthquakes. The log-normal, on the other hand, is not ruled out for any of our data sets except the HTTP connections. In general, we find that it is extremely difficult to tell the difference between log-normal and power-law behavior. Indeed, over realistic ranges of x the two distributions are very close, so it appears unlikely that any test would be able to tell them apart unless we had an extremely large data set. (See the results on synthetic data reported in section 5.)

Finally, for almost a dozen data sets—the forest fires, solar flares, earthquakes, web hits, web links, telephone calls, Internet, email address books, and mammal species—the power law with a cut-off is clearly favored over the pure power law. For surnames the cut-off form is also favored but only weakly, as the p -value is very close to our threshold. For the remaining data sets, the large p -values indicate that there is no statistical reason to prefer the cut-off form over the pure form.

7. Conclusions. The study of power laws spans many disciplines, including physics, biology, engineering, computer science, the earth sciences, economics, political science, sociology, and statistics. Unfortunately, well-founded methods for analyzing power-law data have not yet taken root in all, or even most, of these areas and in many cases hypothesized distributions are not tested rigorously against the data. This leaves open the possibility that conjectured power-law behavior is, in some cases at least, the result of wishful thinking.

In this paper we have argued that the common practice of identifying and quantifying power-law distributions by the approximately straight-line behavior of a histogram on a doubly logarithmic plot should not be trusted: such straight-line behavior is a necessary but by no means sufficient condition for true power-law behavior. Instead, we have presented a statistically principled set of techniques that allow for the

validation and quantification of power laws. Properly applied, these techniques can provide objective evidence for or against the claim that a particular distribution follows a power law. In principle, they could also be extended to other, non-power-law distributions as well, although we have not given such an extension here.

We have applied the methods we describe to a large number of data sets from various fields. For many of these the power-law hypothesis turns out to be, statistically speaking, a reasonable description of the data. That is, the data are compatible with the hypothesis that they are drawn from a power-law distribution, although they are often compatible with other distributions as well, such as log-normal or stretched exponential distributions. In the remaining cases the power-law hypothesis is found to be incompatible with the observed data. In some instances, such as the distribution of earthquakes, the power law is plausible only if one assumes an exponential cut-off that modifies the extreme tail of the distribution.

For some measured quantities, the answers to questions of scientific interest may not rest upon the distribution following a power law perfectly. It may be enough, for example, that a quantity merely have a heavy-tailed distribution. In studies of the Internet, for instance, the distributions of many quantities, such as file sizes, HTTP connections, node degrees, and so forth, have heavy tails and appear visually to follow a power law, but upon more careful analysis it proves impossible to make a strong case for the power-law hypothesis; typically the power-law distribution is not ruled out but competing distributions may offer a better fit to the data. Whether this constitutes a problem for the researcher depends largely on his or her scientific goals. For network engineers, simply quantifying the heavy tail may be enough to allow them to address questions concerning, for instance, future infrastructure needs or the risk of overload from large but rare events. Thus in some cases power-law behavior may not be fundamentally more interesting than any other heavy-tailed distribution. (In such cases, nonparametric estimates of the distribution may be useful, though making such estimates for heavy-tailed data presents special difficulties [36].) If, on the other hand, the goal is, say, to infer plausible mechanisms that might underlie the formation and evolution of Internet structure or traffic patterns, then it may matter greatly whether the observed quantity follows a power law or some other form.

In closing, we echo comments made by Ijiri and Simon [27] more than thirty years ago and similar thoughts expressed more recently by Mitzenmacher [41]. They argue that the characterization of empirical distributions is only a part of the challenge that faces us in explaining the causes and roles of power laws in the sciences. In addition, we also need methods to validate the models that have been proposed to explain those power laws. They also urge that, wherever possible, we consider to what practical purposes these robust and interesting behaviors can be put. We hope that the methods given here will prove useful in all of these endeavors, and that these long-held hopes will at last be fulfilled.

Appendix A. Linear Regression and Power Laws. The most common approach for testing empirical data against a hypothesized power-law distribution is to observe that the power law $p(x) \sim x^{-\alpha}$ implies the linear form

$$(A.1) \quad \log p(x) = \alpha \log x + c.$$

The probability density $p(x)$ can be estimated by constructing a histogram of the data (or alternatively one can construct the CDF by a simple rank ordering of the data) and the resulting function can then be fitted to the linear form by least-squares linear regression. The slope of the fit is interpreted as the estimate $\hat{\alpha}$ of the scaling

parameter. Many standard packages exist that can perform this kind of fitting, provide estimates and standard errors for the slope, and calculate the fraction r^2 of variance accounted for by the fitted line, which is taken as an indicator of the quality of the fit.

Although this procedure appears frequently in the literature, there are several problems with it. As we saw in section 3, the estimates of the slope are subject to systematic and potentially large errors (see Table 2 and Figure 2), but there are a number of other serious problems as well. First, errors are hard to estimate because they are not well described by the usual regression formulas, which are based on assumptions that do not apply in this case. For continuous data, this problem can be exacerbated by the choice of binning scheme used to construct the histogram, which introduces an additional set of free parameters. Second, a fit to a power-law distribution can account for a large fraction of the variance even when the fitted data do not follow a power law, and hence high values of r^2 cannot be taken as evidence in favor of the power-law form. Third, the fits extracted by regression methods usually do not satisfy basic requirements on probability distributions, such as normalization, and hence cannot be correct.

Let us look at each of these objections in a little more detail.

A.1. Calculation of Standard Errors. The ordinary formula for the calculation of the standard error on the slope of a regression line is correct when the assumptions of linear regression hold, which include independent, Gaussian noise in the dependent variable at each value of the independent variable. When fitting to the logarithm of a histogram as in the analysis of power-law data, however, the noise, though independent, is not Gaussian. The noise in the frequency estimates $p(x)$ themselves is Gaussian (actually Poissonian), but the noise in their logarithms is not. (For $\ln p(x)$ to have Gaussian fluctuations, $p(x)$ would have to have log-normal fluctuations, which would violate the central limit theorem.) Thus the formula for the error is inapplicable in this case.

For fits to the CDF the noise in the individual values $P(x)$ is Gaussian (since it is the sum of independent Gaussian variables), but again the noise in the logarithm is not. Furthermore, the assumption of independence now fails, because $P(x) = P(x+1) + p(x)$ and hence adjacent values of the CDF are strongly correlated. Fits to the CDF are, as we showed in section 3, empirically more accurate as a method for determining the scaling parameter α , but this is not because the assumptions of the fit are any more valid. The improvement arises because the statistical fluctuations in the CDF are typically much smaller than those in the PDF. The error on the scaling parameter is thus smaller, but this does not mean that the *estimate* of the error is any better. (In fact, it is typically a gross underestimate because of the failure to account for the correlations.)

A.2. Validation. If our data are truly drawn from a power-law distribution and n is large, then the probability of getting a low r^2 in a straight-line fit is small, so a low value of r^2 can be used to reject the power-law hypothesis. Unfortunately, as we saw in section 4, distributions that are nothing like a power law can appear to follow a power law for small samples and some, like the log-normal, can approximate a power law closely over many orders of magnitude, resulting in high values of r^2 . And even when the fitted distribution approximates a power law quite poorly, it can still account for a significant fraction of the variance, although less than the true power law. Thus, though a low r^2 is informative, in practice we rarely see a low r^2 , regardless of the actual form of the distribution, so that the value of r^2 tells us little. In the terminology of statistical theory, the value of r^2 has very little *power* as a

hypothesis test because the probability of successfully detecting a violation of the power-law assumption is low.

A.3. Regression Lines Are Not Valid Distributions. The CDF must take the value 1 at $x_{\min}$ if the probability distribution above $x_{\min}$ is properly normalized. Ordinary linear regression, however, does not incorporate such constraints and hence, in general, the regression line does not respect them. Similar considerations apply for the PDF, which must integrate to 1 over the range from $x_{\min}$ to ∞ . Standard methods exist to incorporate constraints like these into the regression analysis [65], but they are not used to any significant extent in the literature on power laws.

Appendix B. Maximum Likelihood Estimators for the Power Law. In this appendix we give derivations of the maximum likelihood estimators (MLEs) for the scaling parameter of a power law.

B.1. Continuous Data. In the case of continuous data the MLE for the scaling parameter, first derived (to our knowledge) by Muniruzzaman in 1957 [42], is equivalent to the well-known Hill estimator [24]. Consider the continuous power-law distribution

$$(B.1) \quad p(x) = \frac{\alpha - 1}{x_{\min}} \left(\frac{x}{x_{\min}} \right)^{-\alpha},$$

where α is the scaling parameter and $x_{\min}$ is the minimum value at which power-law behavior holds. Given a data set containing n observations $x_i \geq x_{\min}$, we would like to know the value of α for the power-law model that is most likely to have generated our data. The probability that the data were drawn from the model is proportional to

$$(B.2) \quad p(x | \alpha) = \prod_{i=1}^n \frac{\alpha - 1}{x_{\min}} \left(\frac{x_i}{x_{\min}} \right)^{-\alpha}.$$

This probability is called the *likelihood* of the data given the model. The data are most likely to have been generated by the model with scaling parameter α that maximizes this function. Commonly we actually work with the logarithm $\mathcal{L}$ of the likelihood, which has its maximum in the same place:

$$\begin{aligned} \mathcal{L} = \ln p(x | \alpha) &= \ln \prod_{i=1}^n \frac{\alpha - 1}{x_{\min}} \left(\frac{x_i}{x_{\min}} \right)^{-\alpha} \\ &= \sum_{i=1}^n \left[\ln(\alpha - 1) - \ln x_{\min} - \alpha \ln \frac{x_i}{x_{\min}} \right] \\ (B.3) \quad &= n \ln(\alpha - 1) - n \ln x_{\min} - \alpha \sum_{i=1}^n \ln \frac{x_i}{x_{\min}}. \end{aligned}$$

Setting $\partial \mathcal{L} / \partial \alpha = 0$ and solving for α , we obtain the following MLE for the scaling parameter:

$$(B.4) \quad \hat{\alpha} = 1 + n \left[\sum_{i=1}^n \ln \frac{x_i}{x_{\min}} \right]^{-1}.$$

B.2. Formal Results. There are a number of formal results in mathematical statistics that motivate and support the use of the MLE.

THEOREM B.1. *Under mild regularity conditions, if the data are independent, identically-distributed draws from a distribution with parameter α , then as the sample size $n \rightarrow \infty$, $\hat{\alpha} \rightarrow \alpha$ almost surely.*

Proof. See, for instance, [45]. $\square$

PROPOSITION B.2 (see [42]). *The MLE $\hat{\alpha}$ of the continuous power law converges almost surely on the true α .*

Proof. It is easily verified that $\ln(x/x_{\min})$ has an exponential distribution with rate $\alpha - 1$. By the strong law of large numbers, therefore, $\frac{1}{n} \sum_{i=1}^n \ln \frac{x_i}{x_{\min}}$ converges almost surely on the expectation value of $\ln(x/x_{\min})$, which is $(\alpha - 1)^{-1}$. $\square$

THEOREM B.3. *If the MLE is consistent and there exists an interval $(\alpha - \epsilon, \alpha + \epsilon)$ around the true parameter value α where, for any α_1, α_2 in that interval,*

$$(B.5) \quad \frac{\partial^3 \mathcal{L}(\alpha_1)/\partial \alpha^3}{\partial^2 \mathcal{L}(\alpha_2)/\partial \alpha^2}$$

is bounded for all x , then asymptotically $\hat{\alpha}$ has a Gaussian distribution centered on α , whose variance is $1/nI(\alpha)$, where

$$(B.6) \quad I(\alpha) = -\mathbf{E} \left[\frac{\partial^2 \log p(X|\alpha)}{\partial \alpha^2} \right],$$

which is called the Fisher information at α . Moreover, $\partial^2 \mathcal{L}(\hat{\alpha})/\partial \alpha^2 \rightarrow I(\alpha)$.

Proof. For the quoted version of this result, see [7, Chapter 3]. The first version of a proof of the asymptotic Gaussian distribution of the MLE and its relation to the Fisher information may be found in [18]. $\square$

PROPOSITION B.4 (see [42]). *The MLE of the continuous power law is asymptotically Gaussian, with variance $(\alpha - 1)^2/n$.*

Proof. The proof follows by application of the preceding theorem. Simple calculation shows that $\partial^2 \log \mathcal{L}(\alpha)/\partial \alpha^2 = -n(\alpha - 1)^{-2}$ and $\partial^3 \log \mathcal{L}(\alpha)/\partial \alpha^3 = 2n(\alpha - 1)^{-3}$, so that the ratio in question is $2(\alpha_2 - 1)^2/(\alpha_1 - 1)^3$. Since $\alpha > 1$, this ratio is bounded on any sufficiently small interval around any α and the hypotheses of the theorem are satisfied. $\square$

A further standard result, the *Cramér–Rao inequality*, asserts that for any unbiased estimator of α , the variance is at least $1/nI(\alpha)$. (See [12, section 32.3] or, for an elementary proof, [45].) The MLE is said to be *asymptotically efficient*, since it attains this lower bound.

Proposition B.4 yields approximate standard errors and Gaussian confidence intervals for $\hat{\alpha}$, becoming exact as n becomes large. Corrections depend on how $x_{\min}$ is estimated and on the resulting coupling between that estimate and $\hat{\alpha}$. As the corrections are $O(1/n)$, however, while the leading terms are $O(1/\sqrt{n})$, we have neglected them in the main text. The corrections can be deduced from the “sampling distribution” of $\hat{\alpha}$, i.e., the distribution of deviations from α due to finite-sample fluctuations. (See [12] or [63] for introductions to sampling distributions.) In general, the sampling distribution is hard to obtain analytically, but it can be found by bootstrapping [63, 16]. An important exception is when $x_{\min}$ is either known a priori or simply chosen by fiat (as in the Hill estimator). Starting from the distribution of $\ln x$, it is then easy to show that $(\hat{\alpha} - 1)/n$ has an inverse gamma distribution with shape parameter n and scale parameter $\alpha - 1$. This implies [30] that $\hat{\alpha}$ has a mean

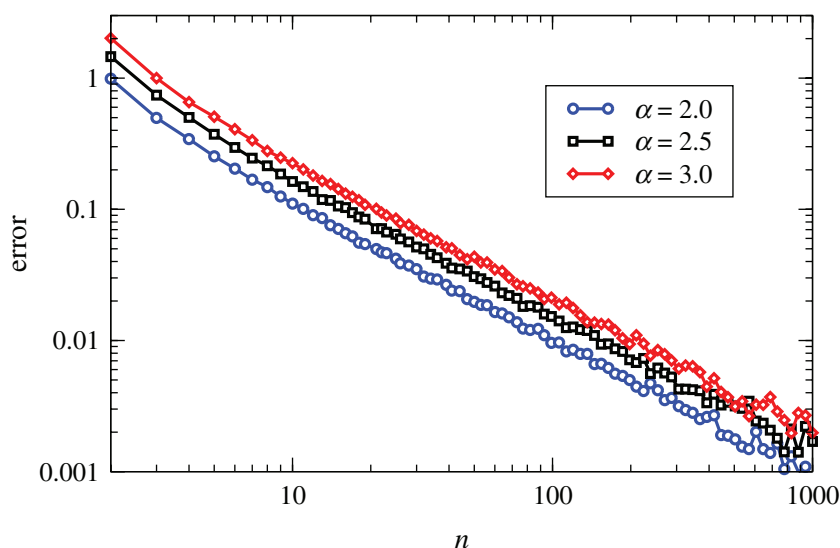


Fig. 10 The error on the estimated scaling parameter $\hat{\alpha}$ from sample size effects for continuous data (similar results hold for the discrete case) for $\alpha = 2, 2.5$, and 3 (for 100 repetitions), as a function of sample size. The average error decays as $O(n^{-1})$ and becomes smaller than 1% of the value of α when $n \gtrsim 50$.

of $(n\alpha - 1)/(n - 1)$ and a standard deviation of $n(\alpha - 1)/(n - 1)\sqrt{n - 2}$, differing, as promised, from the large- n values by $O(1/n)$; see Figure 10.

B.3. Discrete Data. We define the power-law distribution over an integer variable by

$$(B.7) \quad p(x) = \frac{x^{-\alpha}}{\zeta(\alpha, x_{\min})},$$

where $\zeta(\alpha, x_{\min})$ is the generalized or Hurwitz zeta function. For the case $x_{\min} = 1$, Seal [51] and, more recently, Goldstein, Morris, and Yen [19] derived the MLE. One can also derive an estimator for the more general case as follows.

Following an argument similar to the one we gave for the continuous power law, we can write down the log-likelihood function

$$(B.8) \quad \mathcal{L} = \ln \prod_{i=1}^n \frac{x_i^{-\alpha}}{\zeta(\alpha, x_{\min})} = -n \ln \zeta(\alpha, x_{\min}) - \alpha \sum_{i=1}^n \ln x_i.$$

Setting $\partial \mathcal{L} / \partial \alpha = 0$ we then find

$$(B.9) \quad \frac{-n}{\zeta(\alpha, x_{\min})} \frac{\partial}{\partial \alpha} \zeta(\alpha, x_{\min}) - \sum_{i=1}^n \ln x_i = 0.$$

Thus, the MLE $\hat{\alpha}$ for the scaling parameter is the solution of

$$(B.10) \quad \frac{\zeta'(\hat{\alpha}, x_{\min})}{\zeta(\hat{\alpha}, x_{\min})} = -\frac{1}{n} \sum_{i=1}^n \ln x_i.$$

This equation can be solved numerically in a straightforward manner. Alternatively, one can directly maximize the log-likelihood function itself, (B.8).

The consistency and asymptotic efficiency of the MLE for the discrete power law can be proved by applying Theorems B.1 and B.3. As the calculations involved are long and messy, however, we omit them here. Brave readers can consult [6] for the details.

Equation (B.10) is somewhat cumbersome. If $x_{\min}$ is moderately large, then a reasonable figure for α can be estimated using the much more convenient approximate formula derived in the next section.

B.4. Approximate Estimator for the Scaling Parameter of the Discrete Power

Law. Given a differentiable function $f(x)$, with indefinite integral $F(x)$ such that $F'(x) = f(x)$,

$$\begin{aligned} \int_{x-\frac{1}{2}}^{x+\frac{1}{2}} f(t) dt &= F\left(x + \frac{1}{2}\right) - F\left(x - \frac{1}{2}\right) \\ &= \left[F(x) + \frac{1}{2}F'(x) + \frac{1}{8}F''(x) + \frac{1}{48}F'''(x)\right] \\ &\quad - \left[F(x) - \frac{1}{2}F'(x) + \frac{1}{8}F''(x) - \frac{1}{48}F'''(x)\right] + \cdots \\ (B.11) \quad &= f(x) + \frac{1}{24}f''(x) + \cdots. \end{aligned}$$

Summing over integer x , we then get

$$(B.12) \quad \int_{x_{\min}-\frac{1}{2}}^{\infty} f(t) dt = \sum_{x=x_{\min}}^{\infty} f(x) + \frac{1}{24} \sum_{x=x_{\min}}^{\infty} f''(x) + \cdots.$$

For instance, if $f(x) = x^{-\alpha}$ for some constant α , then we have

$$\begin{aligned} \int_{x_{\min}-\frac{1}{2}}^{\infty} t^{-\alpha} dt &= \frac{(x_{\min} - \frac{1}{2})^{-\alpha+1}}{\alpha - 1} \\ &= \sum_{x=x_{\min}}^{\infty} x^{-\alpha} + \frac{\alpha(\alpha+1)}{24} \sum_{x=x_{\min}}^{\infty} x^{-\alpha-2} + \cdots \\ (B.13) \quad &= \zeta(\alpha, x_{\min})[1 + O(x_{\min}^{-2})], \end{aligned}$$

where we have made use of the fact that $x^{-2} \leq x_{\min}^{-2}$ for all terms in the second sum. Thus,

$$(B.14) \quad \zeta(\alpha, x_{\min}) = \frac{(x_{\min} - \frac{1}{2})^{-\alpha+1}}{\alpha - 1} [1 + O(x_{\min}^{-2})].$$

Differentiating this expression with respect to α , we also have

$$(B.15) \quad \zeta'(\alpha, x_{\min}) = -\frac{(x_{\min} - \frac{1}{2})^{-\alpha+1}}{\alpha - 1} \left[\frac{1}{\alpha - 1} + \ln(x_{\min} - \frac{1}{2}) \right] [1 + O(x_{\min}^{-2})].$$

We can use these expressions to derive an approximation to the MLE for the scaling parameter α of the discrete power law, (B.10), valid when $x_{\min}$ is large. The ratio of zeta functions in (B.10) becomes

$$(B.16) \quad \frac{\zeta'(\hat{\alpha}, x_{\min})}{\zeta(\hat{\alpha}, x_{\min})} = -\left[\frac{1}{\hat{\alpha} - 1} + \ln(x_{\min} - \frac{1}{2}) \right] [1 + O(x_{\min}^{-2})],$$

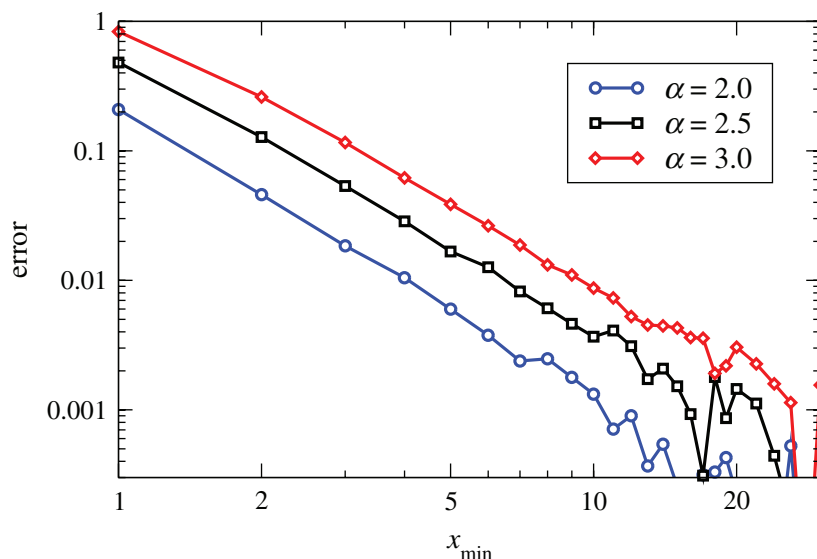


Fig. 11 The error on the estimated scaling parameter $\hat{\alpha}$ that arises from using the approximate MLE for discrete data, (3.7), for $\alpha = 2, 2.5$, and 3 (for 1000 repetitions), as a function of $x_{\min}$. The average error decays as $O(x_{\min}^{-2})$ and becomes smaller than 1% of the value of α when $x_{\min} \gtrsim 6$.

and, neglecting quantities of order $x_{\min}^{-2}$ by comparison with quantities of order 1, we have

$$(B.17) \quad \hat{\alpha} \simeq 1 + n \left[\sum_{i=1}^n \ln \frac{x_i}{x_{\min} - \frac{1}{2}} \right]^{-1},$$

which is in fact identical to the MLE for the continuous case except for the $-\frac{1}{2}$ in the denominator.

Numerical comparisons of (B.17) to the exact discrete MLE, (B.10), show that (B.17) is a good approximation when $x_{\min} \gtrsim 6$ —see Figure 11.

Appendix C. Likelihood Ratio Tests. Consider two different candidate distributions with PDFs $p_1(x)$ and $p_2(x)$. The likelihoods of a given data set within the two distributions are

$$(C.1) \quad L_1 = \prod_{i=1}^n p_1(x_i), \quad L_2 = \prod_{i=1}^n p_2(x_i),$$

and the ratio of the likelihoods is

$$(C.2) \quad R = \frac{L_1}{L_2} = \prod_{i=1}^n \frac{p_1(x_i)}{p_2(x_i)}.$$

Taking logs, the log-likelihood ratio is

$$(C.3) \quad \mathcal{R} = \sum_{i=1}^n [\ln p_1(x_i) - \ln p_2(x_i)] = \sum_{i=1}^n [\ell_i^{(1)} - \ell_i^{(2)}],$$

where $\ell_i^{(j)} = \ln p_j(x_i)$ can be thought of as the log-likelihood for a single measurement x_i within distribution j .

But since, by hypothesis, the x_i are independent, so are the differences $\ell_i^{(1)} - \ell_i^{(2)}$, and hence, by the central limit theorem, their sum $\mathcal{R}$ becomes normally distributed as n becomes large, with expected variance $n\sigma^2$ where σ^2 is the expected variance of a single term. In practice, we don't know the expected variance of a single term, but we can approximate it in the usual way by the variance of the data:

$$(C.4) \quad \sigma^2 = \frac{1}{n} \sum_{i=1}^n \left[(\ell_i^{(1)} - \ell_i^{(2)}) - (\bar{\ell}^{(1)} - \bar{\ell}^{(2)}) \right]^2,$$

with

$$(C.5) \quad \bar{\ell}^{(1)} = \frac{1}{n} \sum_{i=1}^n \ell_i^{(1)}, \quad \bar{\ell}^{(2)} = \frac{1}{n} \sum_{i=1}^n \ell_i^{(2)}.$$

Now suppose we are worried that the true expectation value of the log-likelihood ratio is in fact zero, so that the observed sign of $\mathcal{R}$ is purely a product of the fluctuations and cannot be trusted as an indicator of which model is preferred. The probability that the measured log-likelihood ratio has a magnitude as large as or larger than the observed value $|\mathcal{R}|$ is given by

$$(C.6) \quad p = \frac{1}{\sqrt{2\pi n\sigma^2}} \left[\int_{-\infty}^{-|\mathcal{R}|} e^{-t^2/2n\sigma^2} dt + \int_{|\mathcal{R}|}^{\infty} e^{-t^2/2n\sigma^2} dt \right] \\ = \operatorname{erfc}(|\mathcal{R}|/\sqrt{2n\sigma^2}),$$

where σ is given by (C.4) and

$$(C.7) \quad \operatorname{erfc}(z) = 1 - \operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_z^{\infty} e^{-t^2} dt$$

is the complementary Gaussian error function (a function widely available in scientific computing libraries and numerical analysis programs).

This p -value gives us an estimate of the probability that we measured a given value of $\mathcal{R}$ when the true value of $\mathcal{R}$ is close to zero (and hence is unreliable as a guide to which model is favored). If p is small (say, $p < 0.1$), then our value for $\mathcal{R}$ is unlikely to be a chance result and hence its sign can probably be trusted as an indicator of which model is the better fit to the data. (It does not, however, mean that the model is a *good* fit, only that it is better than the alternative.) If, on the other hand, p is large, then the likelihood ratio test is inadequate to discriminate between the distributions in question.¹²

The rigorous proof of these results involves some subtleties that we have glossed over in our description. In particular, the distributions that we are dealing with are in our case fixed by fitting to the same data that are the basis for the likelihood ratio test and this introduces correlations between the data and the log-likelihoods that must be treated with care. However, Vuong [62] has shown that the results above

¹²Note that, if we are interested in confirming or denying the power-law hypothesis, then a small p -value is “good” in the likelihood ratio test—it tells us whether the test’s results are trustworthy—whereas it is “bad” in the case of the KS test, where it tells us that our model is a poor fit to the data.

do hold even in this case, provided p_1 and p_2 come from distinct, nonnested families of distributions and the estimation is done by maximizing the likelihood within each family. (There are also some additional technical conditions on the models, but they hold for all the models considered here.)

C.1. Nested Hypotheses. When the families of distributions considered are nested, as described in section 5.2, and the true distribution lies in the smaller family, the best fits to both families converge to the true distribution as n becomes large. This means that the individual differences $\ell_i^{(1)} - \ell_i^{(2)}$ in (C.3) each converge to zero, as does their variance σ^2 . Consequently, the ratio $|\mathcal{R}|/\sigma$ appearing in the expression for the p -value tends to $0/0$, and its distribution does not obey the simple central limit theorem argument given above. A more refined analysis, using a kind of probabilistic version of L'Hôpital's rule, shows that in fact $\mathcal{R}$ adopts a chi-squared distribution as n becomes large [67]. One can use this result to calculate a correct p -value giving the probability that the log-likelihood ratio takes the observed value or worse, if the true distribution falls in the smaller family. If this p -value is small, then the smaller family can be ruled out. If not, then the best we can say is that there is no evidence that the larger family is needed to fit to the data, although neither can it be ruled out. For a more detailed discussion of this special case, see, for instance, [62].

Appendix D. Generating Power-Law Distributed Random Numbers. It is often the case in statistical studies of probability distributions that we wish to generate random numbers with a given distribution. For instance, in this paper we have used independent random numbers drawn from power-law distributions to test how well our fitting procedures can estimate parameters such as α and $x_{\min}$. How should we generate such numbers? There are a variety of possible methods, but perhaps the simplest and most elegant is the *transformation method* [46]. The method can be applied to both continuous and discrete distributions; we describe both variants in turn in this section.

Suppose $p(x)$ is a continuous probability density from which we wish to draw random reals $x \geq x_{\min}$. Typically we will have a source of random reals r uniformly distributed in the interval $0 \leq r < 1$, generated by any of a large variety of standard pseudo-random number generators. The probability densities $p(x)$ and $p(r)$ are related by

$$(D.1) \quad p(x) = p(r) \frac{dr}{dx} = \frac{dr}{dx},$$

where the second equality follows because $p(r) = 1$ over the interval from 0 to 1. Integrating both sides with respect to x , we then get

$$(D.2) \quad P(x) = \int_x^\infty p(x') dx' = \int_r^1 dr' = 1 - r$$

or, equivalently,

$$(D.3) \quad x = P^{-1}(1 - r),$$

where P^{-1} indicates the functional inverse of the CDF P . For the case of the power law, $P(x)$ is given by (2.6) and we find that

$$(D.4) \quad x = x_{\min}(1 - r)^{-1/(\alpha-1)},$$

Table 6 Formulas for generating random numbers x drawn from continuous distributions, given a source of uniform random numbers r in the range $0 \leq r < 1$. For the case of the log-normal, there is no simple closed-form expression for generating a single random number, but the expressions given will generate two independent log-normally distributed random numbers x_1, x_2 , given two uniform numbers r_1, r_2 as input. For the case of the power law with cut-off there is also no closed-form expression, but one can generate an exponentially distributed random number using the formula above and then accept or reject it with probability p or $1 - p$, respectively, where $p = (x/x_{\min})^{-\alpha}$. Repeating the process until a number is accepted then gives an x with the appropriate distribution.

Name	Random numbers
Power law	$x = x_{\min}(1 - r)^{-1/(\alpha-1)}$
Exponential	$x = x_{\min} - \frac{1}{\lambda} \ln(1 - r)$
Stretched exponential	$x = \left[x_{\min}^{\beta} - \frac{1}{\lambda} \ln(1 - r) \right]^{1/\beta}$
Log-normal	$\rho = \sqrt{-2\sigma^2 \ln(1 - r_1)}, \theta = 2\pi r_2$ $x_1 = \exp(\rho \sin \theta), x_2 = \exp(\rho \cos \theta)$
Power law with cut-off	see caption

which can be implemented in straightforward fashion in most computer languages.

The transformation method can also be used to generate random numbers from many other distributions, though not all, since in some cases there is no closed form for the functional inverse of the CDF. Table 6 lists the equivalent of (D.4) for a number of the distributions considered in this paper.

For a discrete power law, the equivalent of (D.2) is

$$(D.5) \quad P(x) = \sum_{x'=x}^{\infty} p(x') = 1 - r.$$

Unfortunately, $P(x)$ is given by (2.7), which cannot be inverted in closed form, so we cannot write a direct expression equivalent to (D.4) for the discrete case. Instead, we typically solve (D.5) numerically by a combination of “doubling up” and binary search [46]. That is, for a given random number r , we first bracket a solution x to the equation by the following steps:

```

 $x_2 \leftarrow x_{\min}$ 
repeat
   $x_1 \leftarrow x_2$ 
   $x_2 \leftarrow 2x_1$ 
until  $P(x_2) < 1 - r$ ,
```

where $\leftarrow$ indicates assignment. In plain English, this code snippet tests whether $r \in [x, 2x)$, starting with $x = x_{\min}$ and doubling repeatedly until the condition is met. The end result is a range of x in which r is known to fall. We then pinpoint the solution within that range by binary search. We need only continue the binary search until the value of x is narrowed down to $k \leq x < k + 1$ for some integer k ; then we discard the noninteger part and the result is a power-law distributed random integer. The generalized zeta functions needed to evaluate $P(x)$ from (2.7) are typically calculated using special functions from standard scientific libraries. These functions can be slow, however, so for cases where speed is important, such as cases where we wish

Table 7 CDFs of discrete and continuous power-law distributions with $x_{\min} = 5$ and $\alpha = 2.5$. The second and fourth columns show the theoretical values of the CDFs for the two distributions, while the third and fifth columns show the CDFs for sets of 100 000 random numbers generated from the same distributions using the transformation technique described in the text. The final column shows the CDF for 100 000 numbers generated using the continuous approximation to the discrete distribution, (D.6).

x	Continuous		Discrete		
	Theory	Generated	Theory	Generated	Approx.
5	1.000	1.000	1.000	1.000	1.000
6	0.761	0.761	0.742	0.740	0.738
7	0.604	0.603	0.578	0.578	0.573
8	0.494	0.493	0.467	0.466	0.463
9	0.414	0.413	0.387	0.385	0.384
10	0.354	0.352	0.328	0.325	0.325
15	0.192	0.192	0.174	0.172	0.173
20	0.125	0.124	0.112	0.110	0.110
50	0.032	0.032	0.028	0.027	0.027
100	0.011	0.011	0.010	0.010	0.009

to generate very many random numbers, it may be worthwhile to store the first few thousand values of the zeta function in an array ahead of time to avoid recalculating them frequently. Only the values for smaller x are worth precalculating in this fashion, however, since those in the tail are needed only rarely.

If great accuracy is not needed, it is also possible to approximate the discrete power law by a continuous one. The approximation has to be done in the right way, however, if we are to get good results. Specifically, to generate integers $x \geq x_{\min}$ with an approximate power-law distribution, we first generate continuous power-law distributed reals $y \geq x_{\min} - \frac{1}{2}$ and then round off to the nearest integer $x = \lfloor y + \frac{1}{2} \rfloor$. Employing (D.4), this then gives

$$(D.6) \quad x = \left\lfloor \left(x_{\min} - \frac{1}{2}\right)(1-r)^{-1/(\alpha-1)} + \frac{1}{2} \right\rfloor.$$

The approximation involved in this approach is largest for the smallest value of x , which is by definition $x_{\min}$. For this value the difference between the true power-law distribution, (2.4), and the approximation is given by

$$(D.7) \quad \Delta p = 1 - \left(\frac{x_{\min} + \frac{1}{2}}{x_{\min} - \frac{1}{2}} \right)^{-\alpha+1} - \frac{x_{\min}}{\zeta(\alpha, x_{\min})}.$$

For instance, when $\alpha = 2.5$, this difference corresponds to an error of more than 8% on the probability $p(x)$ for $x_{\min} = 1$, but the error diminishes quickly to less than 1% for $x_{\min} = 5$, and to less than 0.2% for $x_{\min} = 10$. Thus the approximation is in practice a reasonably good one for quite modest values of $x_{\min}$. (Almost all of the data sets considered in section 6, for example, have $x_{\min} > 5$.) For very small values of $x_{\min}$ the true discrete generator should still be used unless large errors can be tolerated. Other approximate approaches for generating integers, such as rounding down (truncating) the value of y , give substantially poorer results and should not be used.

As an example of these techniques, consider continuous and discrete power laws having $\alpha = 2.5$ and $x_{\min} = 5$. Table 7 gives the CDFs for these two distributions, evaluated at integer values of x , along with the corresponding CDFs for three sets of 100 000 random numbers generated using the methods described here. As the table

shows, the agreement between the exact and generated CDFs is good in each case, although there are small differences because of statistical fluctuations. For numbers generated using the continuous approximation to the discrete distribution the errors are somewhat larger than for the exact generators, but still small enough for many practical applications.

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REFERENCES

- [1] J. ABELLO, A. L. BUCHSBAUM, AND J. WESTBROOK, *A functional approach to external graph algorithms*, *Algorithmica*, 32 (2002), pp. 437–458.
- [2] D. ACHLIOPTAS, A. CLAUSET, D. KEMPE, AND C. MOORE, *On the bias of traceroute sampling: or, power-law degree distributions in regular graphs*, *J. ACM*, 56 (4) (2009), article 21.
- [3] L. A. ADAMIC AND B. A. HUBERMAN, *The nature of markets in the World Wide Web*, *Quart. J. Electronic Commerce*, 1 (2000), pp. 5–12.
- [4] R. J. ADLER, R. E. FELDMAN, AND M. S. TAQQU, EDS., *A Practical Guide to Heavy Tails: Statistical Techniques and Applications*, Birkhäuser, Boston, 1998.
- [5] W. AIELLO, F. CHUNG, AND L. LU, *A random graph model for massive graphs*, in *Proceedings of the 32nd Annual ACM Symposium on Theory of Computing*, ACM, New York, 2000, pp. 171–180.
- [6] B. C. ARNOLD, *Pareto Distributions*, International Cooperative Publishing House, Fairland, MD, 1983.
- [7] O. E. BARNDORFF-NIELSEN AND D. R. COX, *Inference and Asymptotics*, Chapman and Hall, London, 1995.
- [8] H. BAUKE, *Parameter estimation for power-law tail distributions by maximum likelihood methods*, *Eur. Phys. J. B*, 58 (2007), pp. 167–173.
- [9] I. BHATTACHARYA AND L. GETOOR, *Collective entity resolution in relational data*, *ACM Trans. Knowledge Discovery from Data*, 1 (1) (2007), article 5.
- [10] A. BRODER, R. KUMAR, F. MAGHOUL, P. RAGHAVAN, S. RAJAGOPALAN, R. STATA, A. TOMKINS, AND J. WIENER, *Graph structure in the web*, *Comput. Networks*, 33 (2000), pp. 309–320.
- [11] A. CLAUSET, M. YOUNG, AND K. S. GLEDITSCH, *On the frequency of severe terrorist events*, *J. Conflict Resolution*, 51 (2007), pp. 58–87.
- [12] H. CRAMÉR, *Mathematical Methods of Statistics*, Almqvist and Wiksells, Uppsala, 1945.
- [13] R. B. D’AGOSTINO AND M. A. STEPHENS, EDS., *Goodness-of-Fit Techniques*, Marcel Dekker, New York, 1986.
- [14] L. DE HANN AND A. FERREIRA, *Extreme Value Theory: An Introduction*, Springer-Verlag, New York, 2006.
- [15] X. DIMITROPOULOS, D. KRIOUKOV, M. FOMENKOV, B. HUFFAKER, Y. HYUN, K. CLAFFY, AND G. RILEY, *AS relationships: Inference and validation*, *ACM SIGCOMM Comput. Comm. Rev.*, 37 (2007), pp. 29–40.
- [16] B. EFRON AND R. J. TIBSHIRANI, *An Introduction to the Bootstrap*, Chapman and Hall, New York, 1993.
- [17] J. D. FARMER, L. GILLEMOT, F. LILLO, S. MIKE, AND A. SEN, *What really causes large price changes?*, *Quant. Finance*, 4 (2004), pp. 383–397.
- [18] R. A. FISHER, *On the mathematical foundations of theoretical statistics*, *Philos. Trans. Roy. Soc. London Ser. A*, 222 (1922), pp. 309–368.
- [19] M. L. GOLDSTEIN, S. A. MORRIS, AND G. G. YEN, *Problems with fitting to the power-law distribution*, *Eur. Phys. J. B*, 41 (2004), pp. 255–258.
- [20] P. D. GRÜN WALD, *The Minimum Description Length Principle*, MIT Press, Cambridge, MA, 2007.
- [21] A. P. HACKETT, *70 Years of Best Sellers, 1895–1965*, R. R. Bowker Company, New York, 1967.

- [22] P. HALL, *On some simple estimates of an exponent of regular variation*, J. Roy. Statist. Soc. Ser. B, 44 (1982), pp. 37–42.
- [23] M. S. HANDCOCK AND J. H. JONES, *Likelihood-based inference for stochastic models of sexual network formation*, Theoret. Population Biol., 65 (2004), pp. 413–422.
- [24] B. M. HILL, *A simple general approach to inference about the tail of a distribution*, Ann. Statist., 3 (1975), pp. 1163–1174.
- [25] P. HOLME, J. KARLIN, AND S. FORREST, *Radial structure of the Internet*, Proc. Roy. Soc. London Ser. A, 463 (2007), pp. 1231–1246.
- [26] M. HUSS AND P. HOLME, *Currency and commodity metabolites: Their identification and relation to the modularity of metabolic networks*, IET Systems Biol., 1 (2007), pp. 280–285.
- [27] Y. IJIRI AND H. A. SIMON, *Skew Distributions and the Sizes of Business Firms*, North-Holland, Amsterdam, 1977.
- [28] T. ITO, K. TASHIRO, S. MUTA, R. OZAWA, T. CHIBA, M. NISHIZAWA, K. YAMAMOTO, S. KUHARA, AND Y. SAKAKI, *Toward a protein–protein interaction map of the budding yeast: A comprehensive system to examine two-hybrid interactions in all possible combinations between the yeast proteins*, Proc. Natl. Acad. Sci. USA, 97 (2000), pp. 1143–1147.
- [29] H. JEFFREYS, *Some tests of significance treated by the theory of probability*, Proc. Cambridge Philos. Soc., 31 (1935), pp. 203–222.
- [30] N. L. JOHNSON, S. KOTZ, AND N. BALAKRISHNAN, *Continuous Univariate Distributions*, John Wiley, New York, 1994.
- [31] R. E. KASS AND A. E. RAFTERY, *Bayes factors*, J. Amer. Statist. Assoc., 90 (1994), pp. 773–795.
- [32] P. L. KRAPIVSKY, S. REDNER, AND F. LEYVAZ, *Connectivity of growing random networks*, Phys. Rev. Lett., 85 (2000), pp. 4629–4632.
- [33] M. KRATZ AND S. I. RESNICK, *The qq-estimator and heavy tails*, Stoch. Models, 12 (1996), pp. 699–724.
- [34] D. J. C. MACKEY, *Information Theory, Inference, and Learning Algorithms*, Cambridge University Press, Cambridge, UK, 2003.
- [35] Y. MALEVERGNE, V. F. PISARENKO, AND D. SORNETTE, *Empirical distributions of log-returns: Between the stretched exponential and the power law?*, Quant. Finance, 5 (2005), pp. 379–401.
- [36] N. M. MARKOVITCH AND U. R. KRIEGER, *Nonparametric estimation of long-tailed density functions and its application to the analysis of World Wide Web traffic*, Performance Eval., 42 (2000), pp. 205–222.
- [37] D. MASON, *Laws of large numbers for sums of extreme values*, Ann. Probab., 10 (1982), pp. 754–764.
- [38] D. G. MAYO, *Error and the Growth of Experimental Knowledge*, University of Chicago Press, Chicago, 1996.
- [39] D. G. MAYO AND D. R. COX, *Frequentist statistics as a theory of inductive inference*, in Optimality: The Second Erich L. Lehmann Symposium, J. Rojo, ed., Institute of Mathematical Statistics, Bethesda, MD, 2006, pp. 77–97.
- [40] M. MITZENMACHER, *A brief history of generative models for power law and lognormal distributions*, Internet Math., 1 (2004), pp. 226–251.
- [41] M. MITZENMACHER, *The future of power law research*, Internet Math., 2 (2006), pp. 525–534.
- [42] A. N. M. MUNIRUZZAMAN, *On measures of location and dispersion and tests of hypotheses in a Pareto population*, Bull. Calcutta Statist. Assoc., 7 (1957), pp. 115–123.
- [43] M. E. J. NEWMAN, *Power laws, Pareto distributions and Zipf’s law*, Contemp. Phys., 46 (2005), pp. 323–351.
- [44] M. E. J. NEWMAN, S. FORREST, AND J. BALTHROP, *Email networks and the spread of computer viruses*, Phys. Rev. E, 66 (2002), article 035101.
- [45] E. J. G. PITMAN, *Some Basic Theory for Statistical Inference*, Chapman and Hall, London, 1979.
- [46] W. H. PRESS, S. A. TEUKOLSKY, W. T. VETTERLING, AND B. P. FLANNERY, *Numerical Recipes in C: The Art of Scientific Computing*, 2nd ed., Cambridge University Press, Cambridge, UK, 1992.
- [47] S. REDNER, *How popular is your paper? An empirical study of the citation distribution*, Eur. Phys. J. B, 4 (1998), pp. 131–134.
- [48] S. I. RESNICK, *Heavy-Tail Phenomena: Probabilistic and Statistical Modeling*, Springer-Verlag, New York, 2006.
- [49] D. C. ROBERTS AND D. L. TURCOTTE, *Fractality and self-organized criticality of wars*, Fractals, 6 (1998), pp. 351–357.
- [50] G. SCHWARZ, *Estimating the dimension of a model*, Ann. Statist., 6 (1978), pp. 461–464.

- [51] H. L. SEAL, *The maximum likelihood fitting of the discrete Pareto law*, J. Inst. Actuaries, 78 (1952), pp. 115–121.
- [52] C. R. SHALIZI, *Dynamics of Bayesian Updating with Dependent Data and Misspecified Models*, Preprint, 2009; available at <http://arxiv.org/abs/0901.1342>.
- [53] M. SMALL AND J. D. SINGER, *Resort to Arms: International and Civil Wars, 1816–1980*, Sage Publications, Beverley Hills, CA, 1982.
- [54] F. A. SMITH, S. K. LYONS, S. K. M. ERNEST, K. E. JONES, D. M. KAUFMAN, T. DAYAN, P. A. MARQUET, J. H. BROWN, AND J. P. HASKELL, *Body mass of late Quaternary mammals*, Ecology, 84 (2003), p. 3403.
- [55] D. SORNETTE, *Critical Phenomena in Natural Sciences*, 2nd ed., Springer-Verlag, Berlin, 2006.
- [56] E. SPRINZAK, S. SATTATH, AND H. MARGALIT, *How reliable are experimental protein-protein interaction data*, J. Molecular Biol., 327 (2003), pp. 919–923.
- [57] S. A. STOEY, G. MICHAILIDIS, AND M. S. TAQQU, *Estimating Heavy-Tail Exponents through Max Self-Similarity*, Preprint, 2006; available at <http://arxiv.org/abs/math.ST/0609163>.
- [58] M. STONE, *Cross-validatory choice and assessment of statistical predictions*, J. Roy. Statist. Soc. Ser. B, 36 (1974), pp. 111–133.
- [59] D. B. STOUFFER, R. D. MALMGREN, AND L. A. N. AMARAL, *Comment on Barabási*, Nature, 435, 207 (2005), Preprint, 2005; available at <http://arxiv.org/abs/physics/0510216>.
- [60] L. TIERNEY, R. E. KASS, AND J. B. KADANE, *Fully exponential Laplace approximations to expectations and variances of nonpositive functions*, J. Amer. Statist. Assoc., 84 (1989), pp. 710–716.
- [61] C. TSALLIS AND M. P. DE ALBUQUERQUE, *Are citations of scientific papers a case of nonextensivity?*, Eur. Phys. J. B, 13 (1999), pp. 777–780.
- [62] Q. H. VUONG, *Likelihood ratio tests for model selection and non-nested hypotheses*, Econometrica, 57 (1989), pp. 307–333.
- [63] L. WASSERMAN, *All of Statistics: A Concise Course in Statistical Inference*, Springer-Verlag, Berlin, 2003.
- [64] L. WASSERMAN, *Frequentist Bayes is objective*, Bayesian Anal., 1 (2006), pp. 451–456.
- [65] S. WEISBERG, *Applied Linear Regression*, 2nd ed., Wiley, New York, 1985.
- [66] M. S. WHEATLAND, *A Bayesian approach to solar flare prediction*, Astrophys. J., 609 (2004), pp. 1134–1139.
- [67] S. S. WILKS, *The large sample distribution of the likelihood ratio for testing composite hypotheses*, Ann. Math. Statist., 9 (1938), pp. 60–62.
- [68] W. WILLINGER AND V. PAXSON, *Where mathematics meets the Internet*, Notices Amer. Math. Soc., 45 (1998), pp. 961–970.
- [69] S.-H. YOOK, Z. N. OLTVAI, AND A.-L. BARABÁSI, *Functional and topological characterization of protein interaction networks*, Proteomics, 4 (2004), pp. 928–942.